A Project Report
on
"N-ONE"
NO ONES ESCAPES

Submitted in partial fulfillment of the requirements



For the award of the degree of
Bachelor of Technology in Computer Science and Engineering

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B. Tech. CSE (Cyber Security)
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Submitted by

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CERTIFICATE

This certifies that the project report entitled "**N-ONE: NO ONES ESCAPES**" submitted by the student as partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering for the period of July - December 2026, at AKS University, Satna, is a bona fide project work carried out by **MAHARSH KEWAT, B2455R10175012,** under my supervision. The supervisor has approved the subject of the project report. This is also to certify that it is his original work and no part of this project is report has been submitted for any other degree/diploma.

All the assistance received during the course of the investigation has been duly acknowledged.

1. I am satisfied that the report presented by **MAHARSH KEWAT** is worthy of consideration for the award of the degree.
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SELF DECLARATION

I hereby declare that the work presented in this project, entitled **"N-ONE: NO ONES ESCAPES ",** towards the partial fulfilment of the requirement for the award of a **Degree in B.Tech** in the Department of Computer Science and Engineering (CYBER SECURITY), **AKS University, Satna (M.P.),** is an authentic record of my own work.

I have not submitted the matter embodied in the project for the award of any other degree or diploma to any other institute or university.



Signature of Candidate

MAHARSH KEWAT

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I am deeply and forever indebted to my parents for their love, support, and encouragement throughout my entire life.

ABSTRACT

In modern security and monitoring operations, traditional surveillance footage often loses its proactive value because systems remain siloed, heavily reliant on manual processing, and strictly reactive in their threat detection mechanisms. Early methods of digital surveillance depended largely on basic motion detection and labor- intensive human observation, making them highly prone to critical human error. Furthermore, traditional facial detection techniques, such as Haar cascades, frequently fail under varied lighting conditions and complex camera angles. Consequently, investigators and security personnel face a pressing need for a unified, intelligent process capable of automatically analyzing video streams, preserving event integrity, and logging tracking data in real-time to prevent security incidents before they escalate.

To address these pervasive vulnerabilities, this project proposes the design and implementation of "N-ONE," a state-of-the-art, AI-driven computer vision surveillance and threat tracking platform. Developed within a Python 3.10+ ecosystem, the proposed framework seamlessly integrates real-time video stream analysis with high-accuracy facial recognition and weapon detection capabilities. By replacing older rule-based algorithms with deep Convolutional Neural Networks (CNNs) and computer vision frameworks like OpenCV, N-ONE adapts dynamically to complex visual environments. The system processes diverse video feeds—ranging from laptop webcams and pre-recorded files to real-time IP camera RTSP streams forced over TCP—ensuring reliable, ordered packet delivery and maintaining strict frame integrity.

The platform operates through a dynamic UI-driven Mode Selector that routes frames into specific artificial intelligence logic workflows without compromising operational efficiency. Mode 1 (Lost Person Search) acts as a target alert system, triggering instant notifications only when a specific registered individual is identified. Mode 2 (Member Attendance) automatically logs known personnel while actively tracking and assigning unique identifiers to unknown individuals, thereby building a persistent database of non-registered subjects. Mode 3 (Threat Detection) utilizes heuristic contour analysis and deep learning object detection architectures to proactively identify hazardous items, such as weapons, providing an immediate digital trail of threats. For identification, the system leverages advanced deep learning backends like Facenet and ArcFace via the DeepFace framework, mapping faces into high-dimensional vector spaces to guarantee highly accurate embedding extraction and minimize false matches.

Crucially, N-ONE establishes a robust data integrity posture suited for modern cybersecurity standards. It enforces strict Role-Based Access Control (RBAC) to segregate administrative and operator privileges, thereby neutralizing the risk of unauthorized database manipulation. To ensure privacy, the framework leverages local deep learning inferencing, preventing sensitive biometric data from being exposed to external cloud networks. Furthermore, the Audit and Logging Module guarantees trail immutability by continuously appending timestamped event data into structured CSV formats in real-time, preventing in-memory data loss during unexpected shutdowns. Ultimately, this project transitions perimeter monitoring from a passive recording utility into a proactive, intelligent assistant. By automating target identification and threat detection, N-ONE provides rapid incident response capabilities and concrete, unalterable audit trails for comprehensive security compliance and verifiable accountability.

TABLE OF CONTENTS

1. INTRODUCTION	**07**
2. OBJECTIVE OF THE PROJECT…	**09**
3. SCOPE OF THE PROJECTS	**11**
4. DEFINITION OF PROBLEM…	**13**
5. BENEFIT OF THE PROJECTS	**15**
6. MAIN MODULE…	**22**
7. TECHNICAL OVERVIEW	**24**
8. SYSTEM CONFIGURATION…	**26**
9. BLOCK DIAGRAM…	**27**
10. DFD DIAGRAM	**28**
11. ER DIAGRAM	**30**
12. PROJECT EXPLANATION…	**31**
13. CODDING (SAMPLE WITH EXPLANATION)	**33**
14. RESULT AND ANALYSIS	**36**
15. SOFTWARE TESTING	**47**
16. SECURITY, PRIVACY, AND ETHICAL CONSIDERATIONS	**51**
17. LIMITATIONS AND FUTURE SCOPE	**52**
18. PREVIEW & SCREENSHOTS	**54**
19. CONCLUSION	**62**
20. BIBLIOGRAPHY / REFERENCES	**63**

INTRODUCTION

Background

Surveillance operators often look at video feeds that show repeated, visual events. A person may need to be found across a camera network. Authorized staff may need to be identified at an entry point. An unusual object may need attention. Manual review can be slow and hard to document in a way. A software system can help by sorting inputs running visual checks, recording decisions and keeping evidence ready for later review.

N-ONE is made to be an integrated dashboard for this help role. N-ONE does not replace a trained operator. Give absolute identity. N-ONE's value is that it joins camera acquisition, profile enrollment face comparison, unknown-person history location-aware logging and a threat-heuristic display in one workflow.

AI surveillance and computer vision

Computer vision systems usually separate image acquisition, detection, representation, comparison or classification and presentation. N-ONE follows this pattern. A frame is captured, faces or possible threat regions are found a branch-specific analysis is. The resulting status is shown in the Streamlit UI. The architecture is purposely modular at the function level even though the deployment is a Python process.

Face recognition context

A face-recognition pipeline is not a face detector. Detection answers where a face-like region is located. Representation turns that region into features. Comparison measures how close the current representation is to an enrolled representation. A threshold turns the distance into a decision. N-ONE's full DeepFace route can use pretrained recognition model interfaces while the active fallback uses detection and HOG/CLAHE features. The report therefore points out the route from the fallback route throughout.

Victim identification problem

Lost Person Search is a one-target verification workflow. The operator selects one registered Victim profile. Supplies a camera location. The implementation limits known matches to that profile and the Victim role. Other faces may be silently tracked in the store but they are not shown as the chosen Victim. This cuts the risk that a different registered person is shown as the search target.

Need for N-ONE

The project tackles five needs:

* One dashboard, for operational modes.

* A controlled registration workflow that stores face crops of arbitrary full images.

* A target-restricted Victim search.

* Persistent unknown-person. Location history.

* A structured audit. Evidence-oriented evaluation workspace.

Project overview

The application starts by creating directories and CSV headers, configuring Streamlit session state rendering the authentication gate and then rendering the authenticated dashboard. An Administrator can register profiles. Change model settings. An Operator can search, monitor inspect the inventory and review logs. The active mode decides whether frames use face processing or the separate threat heuristic branch.

Report organization

Chapters 1-5 set the motivation, objectives, scope, problem and benefits. Chapters 6-13 describe modules, technology, configuration, architecture, data flow, logical data entities, the end-to-end workflow and implementation excerpts. Chapters 14-18 show measured results, tests, security and privacy considerations, limitations, future scope and conclusions. Chapter 19 lists references. Appendices keep evidence plans. Missing academic inputs.

OBJECTIVES OF THE PROJECT

Objective

The main goal is to build an authenticated Streamlit dashboard that helps with face-based Victim Search recognizing people who are registered tracking unknown individuals and monitoring potential threats using video from a local source or a browser.

Functional objectives

- Authenticate users by role: Administrator and Operator.
- Allow staff and victims to be registered using images that are uploaded or captured.
- Ensure one face is visible during registration.
- Save face-only crops with padding using consistent profile IDs.
- Search for one selected victim.
- Compare registered profiles during attendance-style checks.
- Re-identify unknown people locally.
- Record when a victim is seen including the camera location.
- Detect long, weapon-like shapes and warm fire-like areas in the video.
- Show status updates, photos, inventory and logs.

Technical objectives

The system uses OpenCV, NumPy, pandas, Pillow, Streamlit and a DeepFace-compatible adapter. The Administrator can choose models, detectors, metrics and thresholds via the sidebar. The processing width is capped at 1280 pixels, for the OpenCV pipeline. Known and unknown face data are kept in memory.

Security objectives

The application needs four values to run. It compares login credentials using constant-time checking. If secrets are missing the system stays locked. After failed attempts it applies a short lockout. Administrator controls are hidden from Operators. These measures help access but do not make the system a full production-level identity or data protection solution.

Evaluation objectives

The evaluation workspace keeps enrollment data and test data separate. It checks dataset structure records source metadata and compares three recognition models at thresholds. It reports TP, TN, FP, FN, precision, recall, F1, FAR and FRR. When sequence data is missing it clearly marks -frame status as unavailable.

Expected outcomes

The outcome should be a working prototype. It must have reproducible workflows, benchmark evidence backed by source code and clear documentation of what it can and cannot do. It is not expected to deliver identity accuracy always find victims or confirm threats.

SCOPE OF THE PROJECT

Scope

This project scope covers authentication, profile registration, inventory, photo viewing, three operational modes, several video sources, local face storage, tracking of unknown people history of victim sightings audit logs and optional DeepFace-related analytics hooks.

Operational scope

The application scope is meant for a controlled small demonstration environment. One Streamlit process manages the user interface, frame acquisition, in-memory caches and file writes. IP camera access is handled by passing a URL to OpenCV; a central multi-camera orchestration service is not part of this scope.

AI scope

The active fallback within the AI scope uses a CPU-oriented OpenCV/HOG pipeline. The full runtime can expose models such as FaceNet, FaceNet512, ArcFace and others through the DeepFace interface when compatible dependencies are installed. This project scope does not include training a neural network from scratch.

Security scope

Security scope includes loading, role gating, lockout, logout and administrator-only destructive actions. Password hashing, encrypted images encrypted CSVs, TLS configuration, external identity federation and a persistent security event store are outside the security scope.

Dataset scope

The benchmark dataset scope uses five victim identities, 11 enrollment images, 32 victim test images, ten impostor identities represented by 12 images and 60 negative trials formed by comparing each impostor image against five victim identities. Impostor source metadata identifies LFW as the source. This is a limited still-image evaluation scope.

Out-of-scope areas

Staff-specific recognition accuracy, Re-ID accuracy threat-detection accuracy, production camera scalability, multi-frame confirmation metrics, legal compliance certification and institutional academic details are not, within the repository scope.

DEFINITION

Existing surveillance problem

Current surveillance systems split tasks like recording, recognizing, noting incidents and storing evidence into tools. This setup causes a lot of work. It also makes it harder to connect a face to a location, a time and the decisions made during an event.

Recognition failure conditions

Face recognition systems can fail for reasons. Lighting conditions can be bad. The angle of a person's face, blur, distance from the camera or partial faces can cause issues. If someone is partially covered or if multiple people are in the frame the system may not work correctly. Detector. Differences between how a person looked during enrollment and how they appear now add to the problem. N-ONE's registration checks help improve input quality. They cannot fix all the environmental challenges that affect recognition.

False positive and false negative risks

A false positive in Victim Search means an impostor is shown as the Victim. A false negative means the actual Victim is not recognized. The system is designed so that if the neural runtime is not available it treats any fallback identity decision as unsafe for Victim identification. Benchmark reports measure these risks using positives (FP) false negatives (FN) false alarm rate (FAR) false rejection rate (FRR) recall and F1 score.

Unknown-person problem

An unknown person is not a confirmed identity. It is a label given to a person whose face does not match any registered profile. The system may later link this unknown to a saved unknown. Unknown Re-ID asks whether two observations that are both labeled as unknown are likely the person. This is different from Victim Search, which checks if someone matches a known selected identity.

Threat-detection problem

The threat detection module looks for patterns that might look like long objects or warm areas similar to fire. However contours and color masks cannot confirm that something is a gun, knife or fire. The system uses labels like weapon or Possible fire. These are treated as alerts that need human review. The system does not claim certainty, a suggestion based on visual cues.

Formal problem statement

Given an input frame F, a mode M, an optional selected Victim profile V, a location L, a representation function R and a distance threshold τ the system must produce a status that fits the current mode. It must avoid confusing a person, with the selected Victim. In face recognition modes every detected face is. Compared. In Threat Detection Mode face comparison is skipped. Instead the frame is analyzed using contour and HSV-based heuristics. All decisions must be clearly explained along with their evidence limits.

BENEFITS OF THE PROJECT

N-ONE is designed as an integrated AI-assisted surveillance platform that combines subject registration, face-based recognition, target-restricted Victim Search, unknown-person re-identification, threat-oriented visual analysis, location-aware monitoring, structured logging, and experimental model evaluation within a single application. The primary benefit of this architecture is that registration, monitoring, recognition, contextual information, and reviewable records are connected through a common workflow rather than being treated as independent processes. A typical workflow can be represented as Authentication → Profile Selection/Registration → Operational Mode → Video Input → Detection → Analysis → Result → Logging → Operator Review. This integration reduces unnecessary switching between independent tools and provides a consistent operational process. However, N-ONE is an academic and experimental prototype, so the presence of a feature does not automatically establish universal accuracy, production scalability, legal compliance, or guaranteed real-world performance.

Integrated Surveillance Workflow

One of the major benefits of N-ONE is the integration of several surveillance functions into a single Streamlit dashboard. An operator can manage registered subjects, select operational modes, view camera feeds, perform recognition, monitor unknown individuals, review suspicious visual events, and inspect recorded observations within one application environment. This reduces unnecessary tool switching and provides a more consistent workflow. The benefit should primarily be understood as workflow integration rather than a guarantee that the system will operate faster than every alternative implementation.

5.3 Controlled Registration and Profile Management

N-ONE provides a structured registration process for Staff and Victim profiles. During registration, the system expects the image to contain exactly one detectable face, reducing ambiguity about which person should represent the registered identity. The system extracts and stores a padded face crop, providing the recognition pipeline with a more focused reference instead of an arbitrary full image containing unnecessary background or other objects. Profiles also follow a predictable local naming structure, such as Staff_<name>.jpg and Victim_<name>.jpg, which simplifies profile discovery, inventory management, and dataset organization. Controlled enrollment also makes it easier to distinguish registered reference images from later test images during evaluation. However, improved enrollment quality does not eliminate recognition problems caused by lighting, pose, occlusion, camera distance, or other environmental factors.

Target-Restricted Victim Search

Target-restricted Victim Search is one of the key functional benefits of N-ONE. Instead of comparing every detected face against all registered identities, the operator can select a specific Victim before starting the search. The workflow becomes Selected Victim → Camera Input → Face Detection → Target Comparison → Threshold Decision → Victim Result. This creates a clear semantic relationship between the search operation and its intended target. For example, if Victim A is selected and the observed person is Staff B, Staff B should not be presented as "Victim A Found" merely because Staff B is registered. This target restriction reduces unrelated identity information and allows the operator to focus on the selected Victim, recognition result, camera location, timestamp, recognition distance, and sighting history.

Context-Rich and Location-Aware Monitoring

N-ONE can provide more information than a simple match/no-match result. When a selected Victim satisfies the configured recognition conditions, the system can associate the event with information such as Victim name, profile ID, recognition distance, timestamp, camera location, active recognition model or backend, recognition metric, and previous sighting information. This creates a contextual representation based on Who + Where + When + Recognition Evidence. Camera or location information can also connect multiple sightings into a sequence of observations rather than treating them as unrelated events. However, the supplied location value should not be interpreted as independently verified physical geolocation.

Unknown-Person Continuity

Not every person appearing in a surveillance feed will be present in the registered Staff or Victim inventory. N-ONE can maintain context for unmatched observations by assigning a local unknown identifier and attempting to associate sufficiently similar later observations with that identifier. For example, an initial observation may create Unknown_001, and a later sufficiently similar observation may be associated with the same local identifier. This provides basic continuity between observations without claiming that the person's real-world identity is known. Thus, Unknown_001 represents an application-level observation or tracking identity rather than an actual identified person.

Structured Logging and Auditability

N-ONE uses structured CSV-based records for relevant application events, including unknown observations, Victim sightings, and audit information. These records provide a simple and transparent method for reviewing system activity. Because CSV files can be opened using common spreadsheet and data-analysis tools, students, developers, and evaluators can inspect timestamps, locations, IDs, recognition information, and sighting records without requiring a specialized database interface. The records can also be analyzed using Python and Pandas. This is particularly useful in an academic environment because generated evidence can be directly inspected and reused during evaluation. A structured record containing Victim, Location, Timestamp, and Recognition Information provides more persistent information than a temporary dashboard message such as “Victim Found.” However, CSV logging should not be described as an immutable forensic evidence system or enterprise SIEM.

Experimental Model and Threshold Evaluation

An important engineering benefit of N-ONE is the separation between experimental evaluation and live application configuration. Alternative recognition models such as FaceNet, FaceNet512, and ArcFace can be evaluated under controlled conditions without automatically replacing the configuration used by the application. This prevents experimental changes from silently altering operational behavior and makes model testing easier to document. The same evaluation framework can be used to examine different recognition thresholds. A threshold that is too permissive may increase false acceptance, whereas a threshold that is too restrictive may increase false rejection. Therefore, testing multiple thresholds allows the developer to understand how recognition behavior changes with the decision boundary. The purpose is controlled analysis rather than automatic discovery of a universally optimal model or threshold.

Quantitative Recognition and Error Analysis

N-ONE's evaluation framework provides a way to express recognition behavior quantitatively rather than relying only on visual demonstrations. True Positive represents correct acceptance of a genuine target, True Negative represents correct rejection of a non-target, False Positive represents incorrect acceptance of an impostor as the target, and False Negative represents failure to accept the genuine target. From these values, Precision, Recall, F1-score, False Acceptance Rate, and False Rejection Rate can be calculated. These measurements can reveal whether the system misses genuine observations, accepts impostors, becomes more permissive at particular thresholds, becomes more restrictive at others, or requires significant processing time. This provides a stronger basis for technical and research discussion.

Human-in-the-Loop Decision Support

N-ONE follows a human-in-the-loop approach in which AI provides computational assistance while important decisions remain with the authorized operator. When a Victim Search result is generated, the operator can review the captured image, selected Victim profile, recognition distance, timestamp, camera location, sighting history, and model or backend information before interpreting the result. Similarly, suspicious visual detections should be reviewed in context before escalation. This approach recognizes that automated visual systems can produce false positives and false negatives and cannot independently understand every real-world circumstance.

Separate Threat Detection

Threat detection is handled separately from face recognition because the two functions have different objectives. Face recognition attempts to determine whether a detected face corresponds to a relevant registered identity, while threat detection attempts to identify visual patterns associated with the implemented threat heuristics. The threat branch can provide additional monitoring signals, such as possible elongated weapon-like objects or possible warm fire-like regions, which can draw operator attention toward potentially relevant frames. However, these results represent visual indications requiring review rather than confirmed threat classifications. For example, “Possible weapon” should not automatically be interpreted as proof that a weapon has been conclusively detected.

Modular and Flexible Architecture

N-ONE separates major responsibilities into logical modules such as authentication, profile registration, recognition, Victim Search, Unknown Re-ID, threat detection, logging, evaluation, and dashboard presentation. This modularity makes debugging easier because problems can be isolated to specific subsystems. It also supports independent testing, clearer documentation, and future development. Additional recognition models or improved threat algorithms can potentially be introduced without redesigning the entire dashboard.

The system also supports more than one face-processing route. The full recognition environment can use DeepFace-compatible models when the required dependencies are available, while an OpenCV-based fallback route can support applicable functionality when the complete environment cannot be loaded. This provides environmental flexibility, although the fallback should not automatically be considered equivalent to deep-learning-based recognition models.

Local Deployment and Computational Efficiency

N-ONE uses predictable local paths for profiles, logs, unknown observations, and evaluation files. This makes the project easier to inspect and demonstrate in a controlled academic environment without requiring every component to depend on an external cloud service. The complete project can also be copied and examined as a local application, supporting practical reproducibility.

The main OpenCV processing path limits frame width to approximately 1280 pixels, which can help avoid unnecessary processing overhead from very high-resolution input. In-memory known and unknown face caches can also reduce repeated data loading where appropriate. These choices help keep the processing pipeline manageable in a local demonstration environment. However, they do not establish a universal FPS because actual performance depends on hardware, model, detector, resolution, workload, and other operating conditions.

Security and Privacy-Oriented Benefits

The authentication and role-separation mechanisms provide a basic access-control benefit. Administrator controls are separated from Operator workflows, reducing the possibility that every user can directly access sensitive management operations. The application includes mechanisms such as secret-based configuration, constant-time credential comparison, failed-login lockout, logout, and administrator-only destructive operations. These features can improve the security posture of the prototype but do not establish a complete production security architecture.

The controlled registration workflow and local data organization also provide some privacy-oriented benefits. The system uses focused face crops as recognition references, while role separation restricts access to administrative functionality. However, these design choices should not be presented as proof of legal or regulatory compliance. A real deployment would require additional controls for data retention, deletion, access auditing, encryption, privacy governance, and applicable legal requirements.

Academic, Research, and Reproducibility Benefits

N-ONE provides academic value by combining Artificial Intelligence, Computer Vision, Face Recognition, Image Processing, Python Programming, Streamlit Development, Role-Based Access Control, Data Management, Model Evaluation, Security Logging, and Human-in-the-Loop concepts within one implementation. This demonstrates how theoretical concepts can be integrated into a practical application.

The evaluation workspace also supports experimental reproducibility. A benchmark can document the dataset, enrollment and test separation, model, detector, metric, threshold, test conditions, calculated

metrics, and performance measurements. This allows another developer or evaluator to understand how a particular result was obtained. This is particularly important in AI projects because simply stating that "the model worked" does not provide enough information to reproduce an experiment.

N-ONE also promotes evidence-based development by distinguishing assumptions from measurements. Instead of claiming that a particular model is always the best, the evaluation process can examine how each configuration behaves under defined conditions. Similarly, instead of claiming perfect real-time operation, the project can report measured latency together with the hardware and test conditions under which it was obtained.

Maintainability and Future Extensibility

Predictable directories, structured CSV files, modular processing, and explicit configuration make N-ONE easier to maintain. Developers can identify where profiles, unknown observations, Victim sightings, evaluation data, and configuration information are stored and can determine which subsystem is responsible for a particular function.

The architecture also provides a foundation for future improvements such as database-backed storage, improved face-recognition models, GPU acceleration, advanced object detection, better tracking algorithms, multi-camera orchestration, centralized alert management, enterprise authentication, encrypted data storage, advanced audit infrastructure, and larger evaluation datasets. These should remain classified as future scope unless they are implemented and experimentally verified.

Conditions and Limitations

The benefits described in this chapter are conditional on the implementation and operating environment. They do not establish universal face-recognition accuracy, guaranteed Victim discovery, guaranteed threat detection, universal real-time FPS, enterprise-scale camera capacity, complete Staff recognition accuracy, quantitatively validated Unknown Re-ID accuracy, complete legal compliance, complete biometric-data protection, or error-free autonomous surveillance.

Actual system behavior depends on factors including input quality, enrollment quality, lighting, pose, occlusion, camera distance, recognition model, detector, threshold, hardware, dataset composition, and operating environment. Therefore, performance-related claims should always be connected to the specific experimental conditions under which they were measured.

An important distinction should also be maintained between **capability, measurement, and guarantee**. Capability describes what the software has been implemented to do; measurement describes what has been experimentally demonstrated; and guarantee describes what can be expected under all possible real-world conditions. For example, Victim Search is an implemented capability, a benchmark can measure its

behavior, but universal Victim recognition accuracy is not established.

Final Benefit Statement

The overall benefit of N-ONE is the creation of an integrated, evidence-oriented AI-assisted surveillance workflow rather than simply a face-recognition demonstration. The system connects Controlled Registration → Target Selection → Visual Monitoring → Face/Threat Analysis → Recognition or Alert → Location and Timestamp Context → Structured Logging → Human Review → Experimental Evaluation.

This integration makes N-ONE useful as an academic prototype and research-oriented evaluation platform. Its strongest benefits are not claims of perfect recognition or autonomous surveillance, but the combination of controlled workflows, target-specific processing, repeat-observation context, predictable local data handling, measurable recognition evaluation, structured logging, modular architecture, and human review.

At the same time, these benefits do not establish universal recognition accuracy, guaranteed threat classification, measured application-wide FPS, production-scale deployment, or legal compliance. Such claims would require additional datasets, testing, infrastructure, governance, and independent validation. Therefore, N-ONE should be understood as a human-supervised AI-assisted surveillance prototype whose primary value lies in integrating operational workflows and making their results measurable, traceable, and reviewable.

MAIN MODULES

Authentication and RBAC

load_auth_credentials() reads ADMIN_USERNAME, ADMIN_PASSWORD, OPERATOR_USERNAME, and OPERATOR_PASSWORD from the process environment or Streamlit Secrets. If any value is missing, it raises an error and the dashboard remains unavailable. After successful comparison, session state stores authenticated=True and the selected role. Administrators receive registration, model configuration, and deletion controls. Operators receive monitoring and review controls without those destructive or configuration controls.

Profile registration

The Administrator chooses guided multi-angle capture or upload. Guided capture expects front, left, right, up, and down angles. Each image is decoded, face regions are detected, and exactly one face with a minimum size is required. A 25% padded crop is saved under a generated Staff or Victim profile ID. Existing angle files for the same ID are replaced during an update.

Camera ingestion

The browser camera uses an optional WebRTC processor. Local webcams use OpenCV probing across camera indices and Windows backends. Recorded files are written to temp_video_upload.mp4. IP camera URLs are passed to OpenCV. A remote deployment should prefer browser capture because the server process cannot directly access a user's local camera.

Face recognition and matching

The active DeepFace-compatible object returns face regions and embeddings. The application searches the known cache first. In Victim Search, the search is restricted by profile ID and role. In attendance operation, all known profiles are candidates. Unmatched faces are checked against the unknown cache and then registered if no prior unknown matches.

Unknown Re-ID

A new unknown receives unknown_001, unknown_002, and so on. Its crop is saved, its first and last sighting metadata is added to unknown_person_db.csv, and a sighting row is appended. A later match updates the last-known location and timestamp and writes another sighting subject to a two-second write throttle for the same ID/location pair.

Victim Search

The operator selects a Victim and enters a camera location before starting. A genuine match creates the Victim Found event and a dedicated result card with the profile image, ID, location, distance, timestamp, active model/backend/metric, and latest sighting per location. In the OpenCV fallback, visible identity

matching is disabled; the UI can confirm face detection without labeling a Victim.

Threat Detection

Threat mode is exclusive. It does not use FaceNet, FaceNet512, ArcFace, or the cosine threshold. It applies Canny edges, morphology, contour geometry, and a warm-colour HSV mask. Alerts are throttled before audit logging. The output is explicitly possible threat/fire, not verified classification.

Dashboard and logs

The dashboard shows role, registered counts, unknown counts, and total log events. Authenticated users can filter registered faces by All, Staff, and Victim, view photos and metadata, inspect unknown records, and review Victim sighting history.

Table 4 - Project modules

Module	Status	Main output
Authentication/RBAC	Implemented	Role-gated session
Registration	Implemented	Face-only profile images
Victim Search	Implemented with runtime dependency boundary	Target result or no-match
Staff/Attendance	Implemented workflow; Staff benchmark unavailable	Known/unknown status
Unknown Re-ID	Implemented local storage workflow; accuracy unavailable	Unknown ID/history
Threat Detection	Implemented heuristic path	Possible threat/fire alert
WebRTC camera	Optional/conditional	Worker-thread annotation
Evaluation	Implemented evidence workspace	CSV metrics
New neural training	Not implemented	None

TECHNICAL OVERVIEW

Technology stack

Technology	Confirmed role
Python	Application language
Streamlit 1.60.0	Dashboard and session state
OpenCV	Image/video processing, Haar detection, contours, annotations
NumPy	Array and distance calculations
pandas	CSV persistence and tabular display
Pillow	Uploaded-image decoding
DeepFace 0.0.100	Conditional neural face-analysis interface
TensorFlow	Conditional dependency for full neural runtime
streamlit-webrtc 0.77.0	Optional browser camera path on supported Python versions
PyAV	Optional WebRTC frame conversion

DeepFace and model roles

FaceNet, FaceNet512, and ArcFace are representation models used for face comparison when the full runtime is active. RetinaFace is a detector, not an identity model. OpenCV is the source default detector. Threat detection is a separate contour/HSV path and must not be described as FaceNet weapon detection.

Cosine distance

For vectors $a$and b:

$$d_{cos}(a,b) = 1 - \frac{a \cdot b}{|a||b|}$$

N-ONE treats a smaller distance as more similar and accepts a match when the distance is less than or equal to the selected threshold. The default threshold is 0.40, but thresholds are model- and detector-specific.

Fallback representation

OpenCVFaceBackend uses frontal and profile Haar cascades, equalized grayscale, optional upscaling, and overlapping-box deduplication. It extracts a 64x64 grayscale crop, applies CLAHE, computes a HOG descriptor, and normalizes the vector. This is a lightweight fallback, not a trained identity network**.**

Implemented and conditional features

The UI includes hooks for attributes, face extraction, one-to-one verification, and anti-spoofing. Their actual availability depends on the loaded backend. The active fallback implements the core representation/find/verify adapter surface but does not provide evidence that all neural attribute or liveness operations are active in the audited environment.

SYSTEM CONFIGURATION

Software configuration

The base requirements target Python 3.10+ and document support for the OpenCV fallback on Python 3.14. The optional neural requirements target Python 3.10-3.13 because TensorFlow wheels are environment-dependent. The audited benchmark environment was Python 3.12.10 on Windows 11. The benchmark report says RAM was unavailable and GPU was not measured; CPU execution was requested.

Hardware configuration

The repository does not provide a measured application hardware profile. The benchmark provides processor text but no complete deployment specification. Therefore, this report does not claim an application minimum CPU, RAM, GPU, FPS, or camera count. A final university submission should add the actual demonstration machine details.

Network and camera configuration

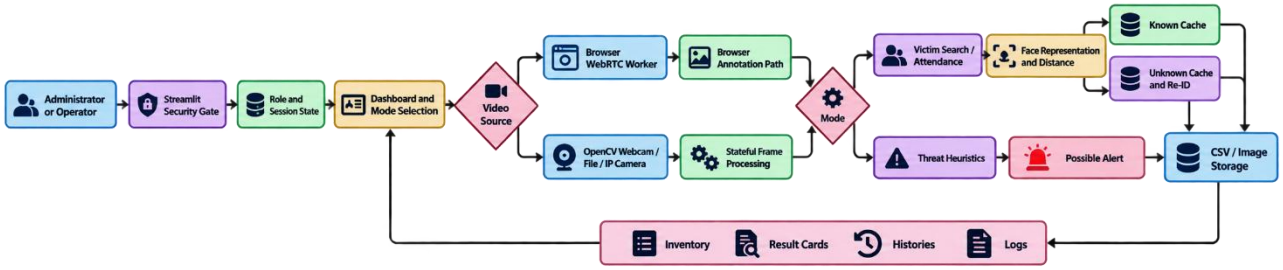
Browser capture requires browser permission and WebRTC dependencies. Host webcam and file/IP-stream processing use OpenCV. RTSP credentials or URLs must not be copied into the academic report.

Configuration table

Parameter	Current source value/status
Recognition model default	Facenet
Detector default	opencv
Metric default	cosine
Threshold default	0.40
Processing width cap	1280 pixels
Victim duplicate suppression	Same profile/location under 60 seconds
Unknown sighting write interval	2 seconds per ID/location
Login attempts	5 before lockout
Login lockout	60 seconds
Application RAM/FPS/GPU	Not measured in current implementation/evaluation

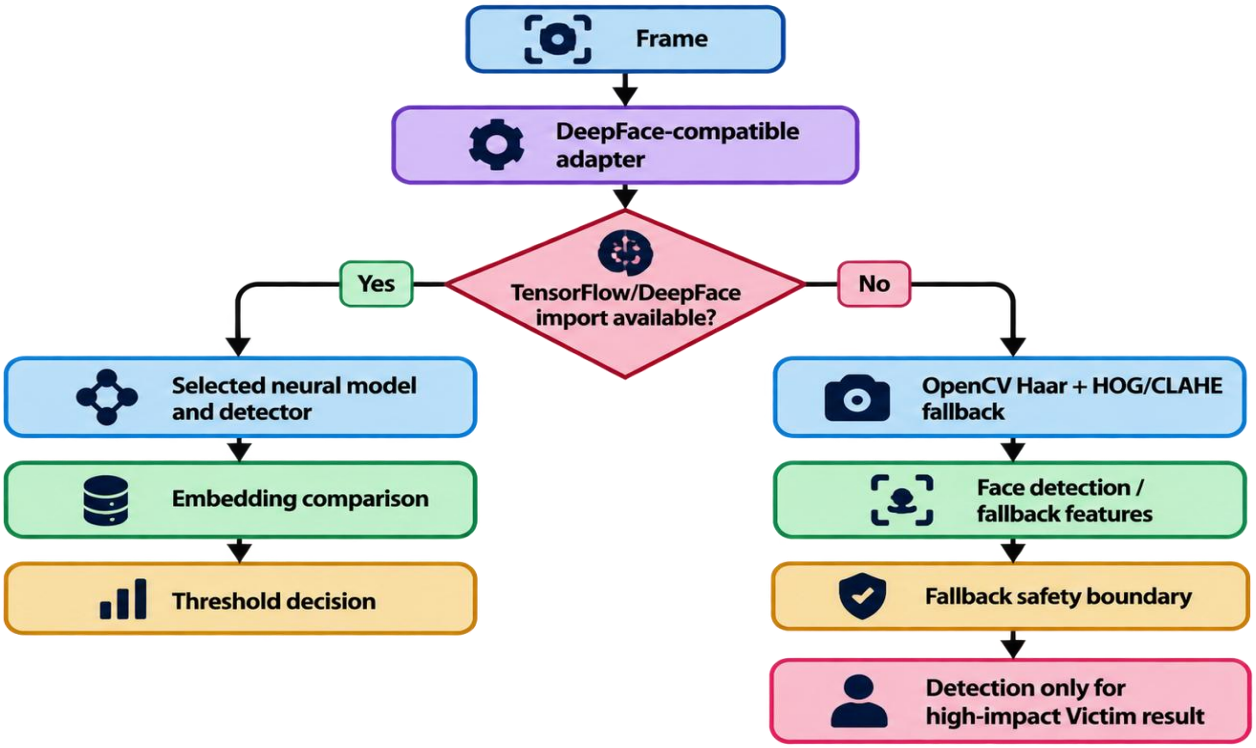
BLOCK DIAGRAM & SYSTEM ARCHITECTURE

Overall architecture



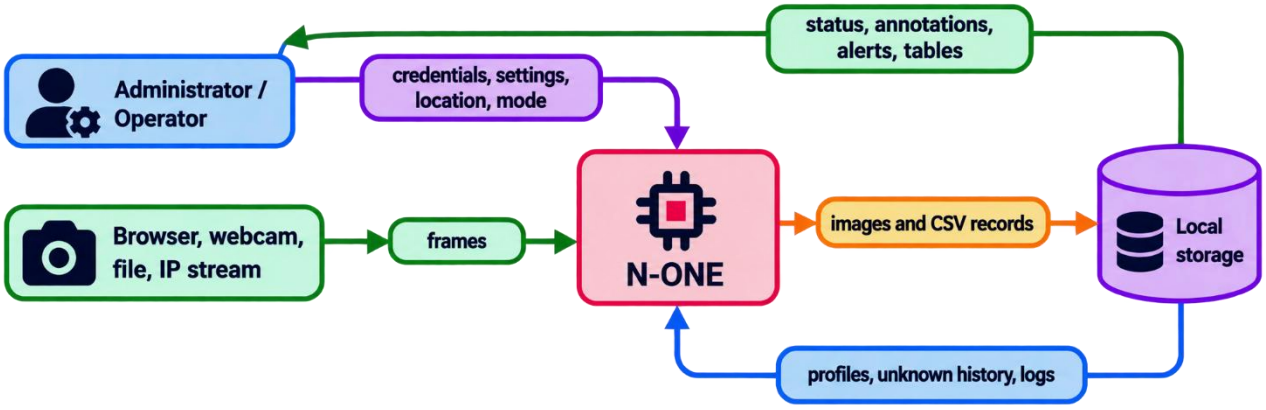
The architecture has a presentation layer, processing layer, and local storage layer. The application is monolithic; function boundaries provide organization but there is no separate service layer or database server.

Conditional AI boundary

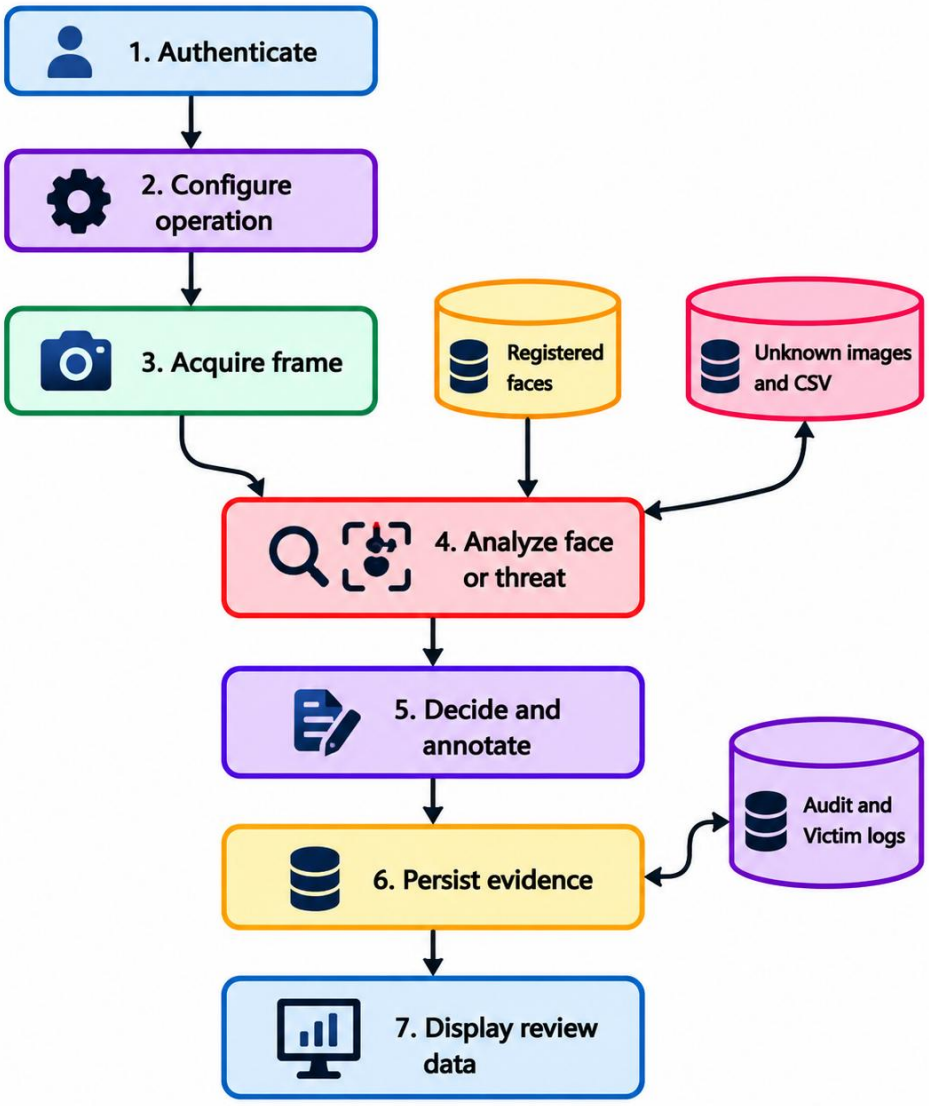


DATA FLOW DIAGRAMS

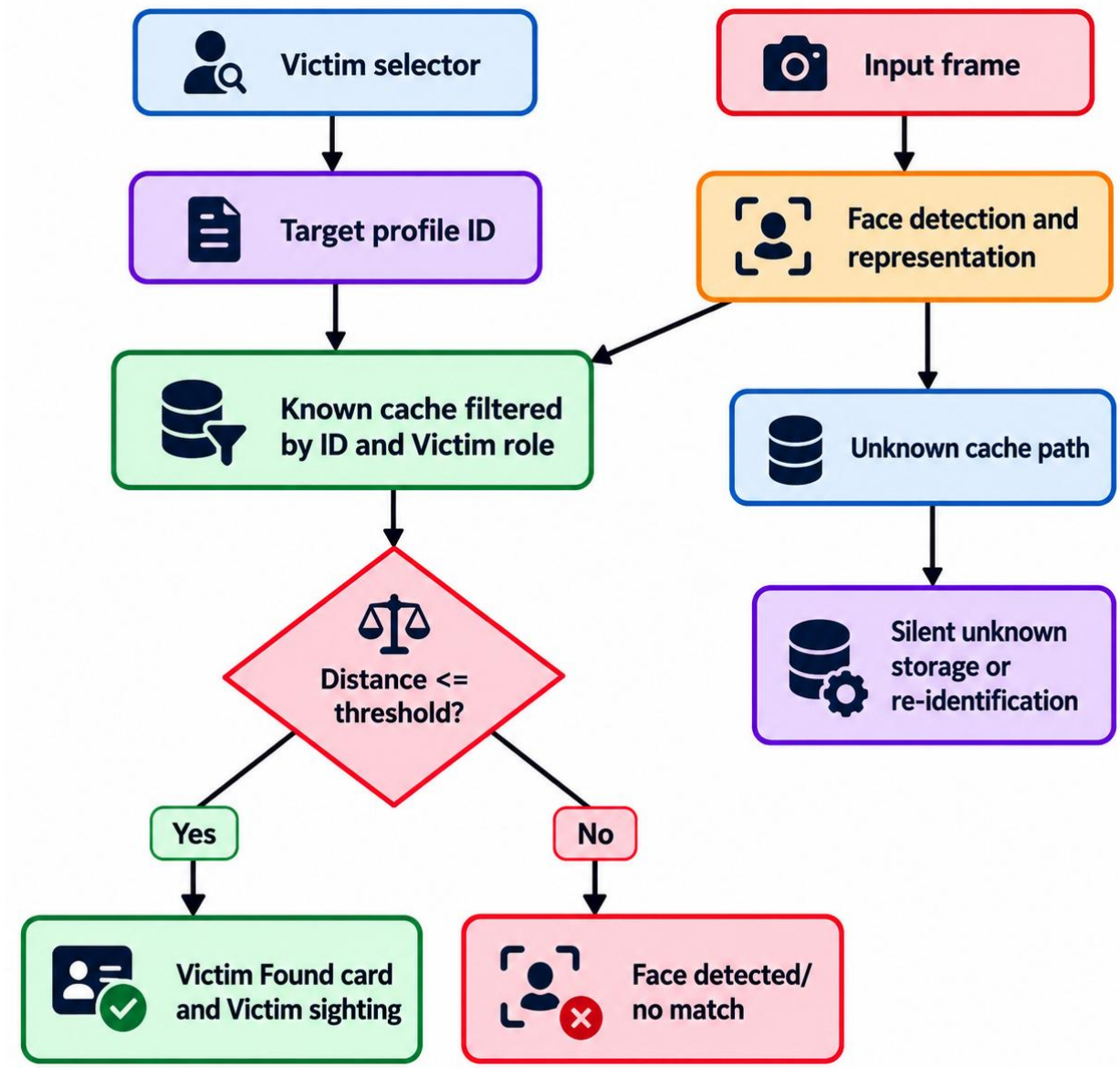
Context DFD



Level-0 DFD



Victim Search DFD



Data stores

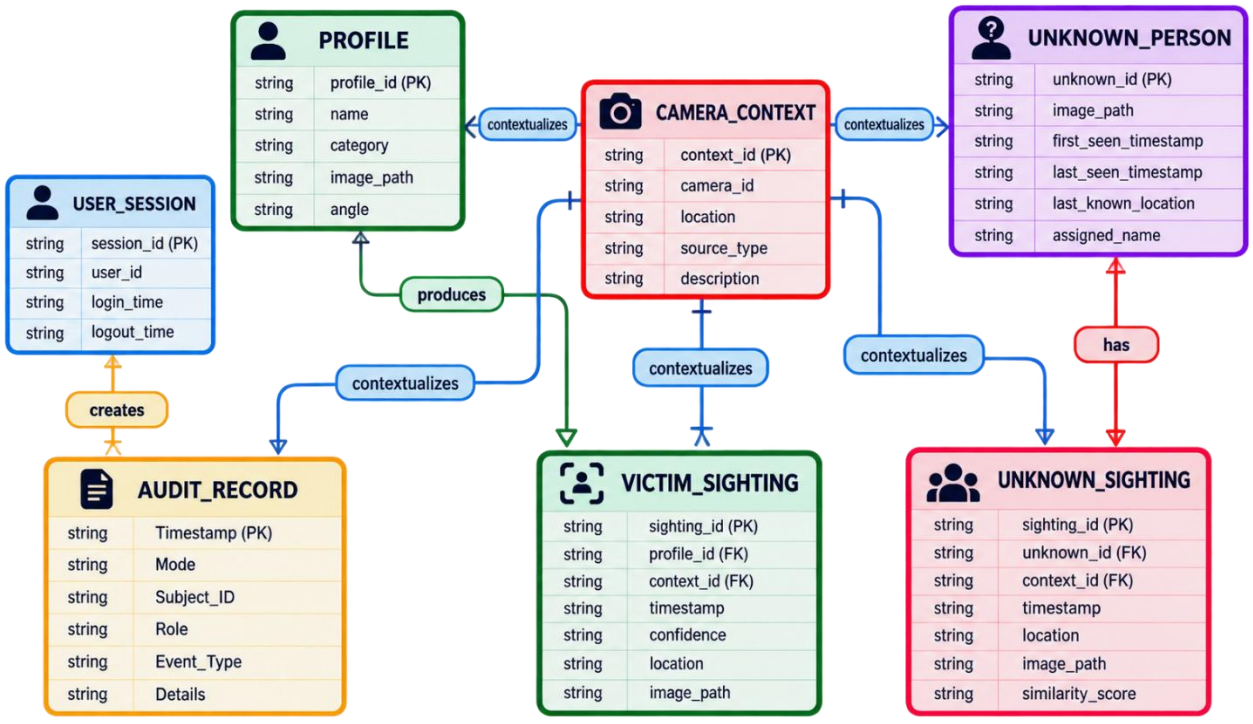
The actual stores are image directories and CSV files. There is no SQL ER implementation. The logical entity model in Chapter 11 is therefore a documentation model derived from the file formats.

LOGICAL ER / DATA MODEL

Logical entities

- User session: role, authentication state, selected settings.
- Profile: profile ID, name, category, image paths, angle.
- Unknown Person: unknown ID, image path, first/last timestamps, last location, optional assigned name.
- Unknown Sighting: sighting ID, unknown ID, timestamp, location.
- Victim Sighting: profile ID, name, timestamp, location.
- Audit Record: timestamp, mode, subject ID, role, event type, details.
- Camera context: operator-entered location associated with an event.

Logical relationship diagram



This is a logical data model, not a claim that the application creates SQL tables or enforces relational constraints.

PROJECT EXPLANATION

Startup

main() initializes directories and log files, sets session defaults, renders the sidebar, loads known encodings for authenticated sessions, and renders the main UI, analytics panel, and log viewer. Missing credentials leave the system locked.

Authentication

The Administrator or Operator enters credentials in the sidebar. The application reads secret values from the environment or Streamlit Secrets, compares values, assigns a role, and reruns the page. After five failures the session is locked for 60 seconds. Logout clears authentication and role state.

Profile enrollment

A registration image is read with Pillow and converted to BGR. Face regions are obtained through the loaded backend. Exactly one region is required. The crop receives padding and is written to the registered directory. In guided mode, five angle files are associated with one profile ID.

Victim Search

The operator selects 1. Lost Person Search, chooses one Victim, enters a camera location, selects a source, and starts surveillance. The stateful OpenCV path processes each frame. In the neural path, the face is compared only with the selected Victim cache entries. A match is recorded, annotated, and shown in a dedicated card. A non-match is not assigned the Victim identity. The fallback path reports that identity matching requires the neural runtime.

Staff Recognition / Attendance

The attendance mode compares detected faces with the unrestricted known cache. A registered match is labeled. If no registered match is found, the system checks the unknown cache, displays a previous unknown ID when found, or saves a new unknown crop and metadata.

The repository does not contain an independent Staff benchmark. Therefore, no Victim benchmark number is reused here. Staff-specific recognition accuracy is **Not measured in the current implementation/evaluation.**

Unknown Re-ID

A locally saved unknown image is treated as a reference representation. A later face is compared with the unknown cache using the selected metric and mode-specific threshold map. A repeat updates the last sighting and appends a log row. The system's unknown ID is an application identifier, not a legal identity.

Threat Detection

The threat mode runs check_weapon_contours() and returns before face processing. Edge contours are filtered by area, aspect ratio, fill ratio, and solidity. A separate HSV mask searches for saturated warm regions. The result can be Possible weapon, Possible fire, or no threat. No FaceNet model or face threshold is involved.

Browser WebRTC path

The browser path uses a worker-thread processor. It annotates frames but deliberately avoids Streamlit state writes, CSV logging, and disk writes in the callback. The latest selected-Victim result is placed in a protected status object and polled by the UI. This behavior differs from the stateful OpenCV loop and should be described separately in an academic demonstration.

Error handling

The application handles missing optional WebRTC imports, missing secrets, invalid image files, no-face and multiple-face registration cases, black frames, unavailable camera sources, missing files, CSV parsing failures, and backend exceptions. Some optional analytics helpers return a fallback result when the backend is unavailable. Those fallbacks must not be described as successful liveness or attribute inference.

CODING & IMPLEMENATATION

The report should include short, readable code listings rather than dumping app.py. The following excerpts are representative source-backed listings.

Authentication secret loading

File: app.py

Function: _read_secret() and load_auth_credentials()

Purpose: Load required credentials without hardcoding them.

```
def _read_secret(name: str) -> str | None:
    environment_value = os.getenv(name)
    if environment_value is not None:
        return environment_value
    try:
        secret_value = st.secrets.get(name)
    except StreamlitSecretNotFoundError:
        return None
    return None if secret_value is None else str(secret_value)
```

The function checks process configuration first and Streamlit Secrets second. It returns no secret value when the configuration source is unavailable. load_auth_credentials() rejects incomplete configuration, which prevents an accidental default-login path.

Distance calculation

```
def calculate_face_distance(face_encoding, reference_encoding, metric="cosine"):
    probe = np.asarray(face_encoding, dtype=np.float64)
    reference = np.asarray(reference_encoding, dtype=np.float64)
    if probe.shape != reference.shape:
        raise ValueError("Face embedding dimensions do not match")
    if not np.all(np.isfinite(probe)) or not np.all(np.isfinite(reference)):
        raise ValueError("Face embedding contains non-finite values")
    if metric == "euclidean":
        return float(np.linalg.norm(probe - reference))
    if metric == "euclidean_l2":
```

```
    return float(np.linalg.norm(probe - reference) /
                 max(np.linalg.norm(probe) + np.linalg.norm(reference), 1e-8))
  return float(1.0 - np.dot(probe, reference) /
               (np.linalg.norm(probe) * np.linalg.norm(reference)))
```

The implementation validates shape and finite values. A zero vector is treated as a non-match in the current source path.

Target-restricted matching

```
known_match = find_face_in_known_cache(
  face_encoding,
  match_threshold,
  profile_id=target_profile_id,
  role="Victim",
  metric=metric,
) if target_profile_id else None
```

The profile ID and role filter are the controlling safeguards for visible Victim Search identity results. A different registered profile cannot satisfy this branch.

Registration validation

```
regions = detect_face_regions(frame)
if len(regions) == 0:
  return None, "No face detected. Face the camera directly with good lighting."
if len(regions) > 1:
  return None, "Multiple faces detected. Keep only one person in the frame."
```

The code then checks the minimum crop size, applies padding, and returns the face-only crop for persistence.

Threat heuristic boundary

```
if aspect_ratio >= 2.5 and fill_ratio >= 0.08 and solidity >= 0.15:
  boxes.append((x, y, w, h, "Possible weapon"))
```

The label is intentionally Possible weapon. The source docstring states that contour analysis cannot prove an object is a gun or knife. That wording is required in the report and operator UI interpretation.

Unknown registration

register_new_unknown() creates a sequential ID, writes the face crop, appends a row to the unknown database, appends the first sighting, and writes an audit event. update_unknown_sighting() updates last-seen metadata and logs later sightings with a short duplicate-write interval.

Implementation limitations

The single-module design makes the prototype easy to run but concentrates UI, storage, and processing responsibilities in app.py. Direct CSV writes can be vulnerable to concurrent-access problems in a multi-worker deployment. The project does not add a database transaction layer, schema migration system, or encrypted storage wrapper.

RESULT & ANALYSIS

Evaluation protocol

The benchmark separates Victim enrollment images from Victim test images. It uses five Victim identities, 11 enrollment images, 32 genuine test images, ten impostor identities represented by 12 images, and 60 negative trials. The detector/backend is RetinaFace where supported, the distance metric is cosine, and the tested threshold range is 0.20 through 0.60.

A genuine trial is a test image of the target Victim. An impostor trial compares an impostor image against a Victim identity. A decision is accepted when the minimum distance to the target's enrollment representations is less than or equal to the threshold.

Threshold-0.40 model comparison

Model	TP	TN	FP	FN	Precision	Recall	F1	FAR	FRR	False Victim matches
FaceNet	21	60	0	11	1.0000	0.65625	0.7924528302	0.0000	0.34375	0
FaceNet512	22	60	0	10	1.0000	0.68750	0.8148148148	0.0000	0.31250	0
ArcFace	23	60	0	9	1.0000	0.71875	0.8363636364	0.0000	0.28125	0

At threshold 0.40, ArcFace produced the strongest measured recall and F1 among these three tested model configurations while recording zero false Victim matches in the available negative trials. The claim is limited to this dataset and protocol. It is not a universal accuracy claim, a safety guarantee, or an automatic production recommendation.

A – Complete Model Comparison CSV

MODEL	STATUS	THRESHOLD	TP	TN	FP	FN	PRECISION	RECALL	F1	FAR	FRR	FALSE_VICTIM_MATCHES	GENUINE_TRIALS	IMPOSTOR_TRIALS	ERROR
Facenet	measured	0.4	21	60	0	11	1.0	0.65625	0.7924528301886793	0.0	0.34375	0	32	60	
Facenet512	measured	0.4	22	60	0	10	1.0	0.6875	0.8148148148148148	0.0	0.3125	0	32	60	
ArcFace	measured	0.4	23	60	0	9	1.0	0.71875	0.8363636363636363	0.0	0.28125	0	32	60	

B – Victim/Genuine Trial-Level Results

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
Facenet	genuine	evaluation\dataset\victims\test\normal\Maharsh\718c68ec86c845cc8e9b605220d6cbba.jpg	Maharsh	Maharsh	0.1279774803765683	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\angle\Maharsh\84f9319e32684b5fa35baf8c309885f1.jpg	Maharsh	Maharsh	0.1430357048126487	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ed6e345cfe0b4c1a91f7715f95cc59e3.jpg	Maharsh	Maharsh	0.3053874733579474 7	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\Maharsh\0d291f45643f4fecb868110f91ab344c.jpg	Maharsh	Maharsh	0.3053874733579474 7	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\Maharsh\f735fa5b36584eb08e9c957942b600e7.jpg	Maharsh	Maharsh	0.1934897055613609 1	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\blur\Maharsh\984206e6d13c4872bb4ef556aa27d5dd.jpg	Maharsh	Maharsh	0.1236541467419934 9	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0d4cfd9b1ea346d49bdab8e7acb5f6a5.jpg	Maharsh	Maharsh	0.4610356760197969	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0b43efbaa6a148a28ecf8fb4e07aa14c.jpg	Maharsh	Maharsh	0.1699109653641470 7	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ba32ad437d5d43128b5e54de15a3d2a2.jpg	Maharsh	Maharsh	0.1699109653641470 7	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\lighting\Maharsh\71f33f70fda54b659e9f434e9700fd86.jpg	Maharsh	Maharsh	0.2321521068942512 5	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\Maharsh\816dd95ad42740eb825d18b167afcc0b.jpg	Maharsh	Maharsh	0.1904267359220378 5	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\angle\Maharsh\6db477703c7a48fb927c05f75db4dc04.jpg	Maharsh	Maharsh	0.1445238379582062 2	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\Maharsh\89d1e95a4a8f4d598a9d3edfcf14489d.jpg	Maharsh	Maharsh	0.2151029562592986 5	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\normal\ravendra\780bb17cf73a4a65a64cf9d19633c9c6.jpg	ravendra	ravendra	0.4494938335942666 5	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\angle\ravendra\a35b0f62251f4773ae0569f71d118ba3.jpg	ravendra	ravendra	0.5163130653102712	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\lighting\ravendra\8b5221af4b654f01973d387baa1e5fce.jpg	ravendra	ravendra	0.5163130653102712	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\distance\ravendra\103c171bcf7340428ccf00f49a5cd24e.jpg	ravendra	ravendra	0.3977272545903191	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\ravendra\143852b7e5764fe480997be906bb3cf7.jpg	ravendra	ravendra	0.4345468080052399	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\blur\ravendra\b82b2d3507cc4ab49d1d20b9c8dc7ef0.jpg	ravendra	ravendra	0.5129089967530409	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\multiple_faces\ravendra\bdf58eda61214032880fdee25686cd32.jpg	ravendra	ravendra	0.4937967882599984	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\normal\ravi\1ff748b0db1f4d9b88b4c5176386658d.jpg	ravi	ravi	0.0297476796334830 15	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\normal\ravi\93686ec7f91c418aa37c28912b134098.jpg	ravi	ravi	0.0374515604110159 4	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\angle\ravi\bd7f690c2bcd4fd39caa74042f0a16d8.jpg	ravi	ravi	0.0374515604110159 4	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\lighting\ravi\efb8397169914bf3aa31ecf01b687dae.jpg	ravi	ravi	0.5464521923023655	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\lighting\ravi\d364e0f3e65940a68528351029fe72db.jpg	ravi	ravi	0.5464521923023655	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\distance\ravi\02badeacfa984bf5b4c05262fb51f548.jpg	ravi	ravi	0.5464521923023655	ok	0.4	False
Facenet	genuine	evaluation\dataset\victims\test\distance\ravi\8db138824f1d42b4a41945886f697eba.jpg	ravi	ravi	0.0283914025174883 95	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\multiple_faces\ravi\041528dd6f1b4e68b65b9040f7309345.jpg	ravi	ravi	0.0374515604110159 4	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\normal\shubham\f34981c1e8ef4e5282426332e6fa0851.jpg	shubham	shubham	0.1974098703908287	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\angle\shubham\fcae35a9f88c4035b7f8da6a44bf1549.jpg	shubham	shubham	0.0219376323567612 72	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\distance\shubham\657bff67ea954cdc9cf4ef242a7bdb87.jpg	shubham	shubham	0.3675375328606832 5	ok	0.4	True
Facenet	genuine	evaluation\dataset\victims\test\blur\shubham\4570bcdc073846029ac8515c481344dd.jpg	shubham	shubham	0.4258286345219888	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\normal\Maharsh\718c68ec86c845cc8e9b605220d6cbba.jpg	Maharsh	Maharsh	0.1278353482077019 3	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\angle\Maharsh\84f9319e32684b5fa35baf8c309885f1.jpg	Maharsh	Maharsh	0.1252851265540761	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ed6e345cfe0b4c1a91f7715f95cc59e3.jpg	Maharsh	Maharsh	0.3460522442370347 3	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\distance\Maharsh\0d291f45643f4fecb868110f91ab344c.jpg	Maharsh	Maharsh	0.3460522442370347 3	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\distance\Maharsh\f735fa5b36584eb08e9c957942b600e7.jpg	Maharsh	Maharsh	0.1919250668197376 4	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\blur\Maharsh\984206e6d13c4872bb4ef556aa27d5dd.jpg	Maharsh	Maharsh	0.1033381983037380 3	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0d4cfd9b1ea346d49bdab8e7acb5f6a5.jpg	Maharsh	Maharsh	0.3647695367358233 5	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0b43efbaa6a148a28ecf8fb4e07aa14c.jpg	Maharsh	Maharsh	0.1223900296904335 5	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ba32ad437d5d43128b5e54de15a3d2a2.jpg	Maharsh	Maharsh	0.1223900296904335 5	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\lighting\Maharsh\71f33f70fda54b659e9f434e9700fd86.jpg	Maharsh	Maharsh	0.1665641019383374 8	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\distance\Maharsh\816dd95ad42740eb825d18b167afcc0b.jpg	Maharsh	Maharsh	0.1621149137124767 7	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\angle\Maharsh\6db477703c7a48fb927c05f75db4dc04.jpg	Maharsh	Maharsh	0.0865616248943415	ok	0.4	True

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
Facenet512	genuine	evaluation\dataset\victims\test\distance\Maharsh\89d1e95a4a8f4d598a9d3edfcf14489d.jpg	Maharsh	Maharsh	0.14746598811127032	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\normal\ravendra\780bb17cf73a4a65a64cf9d19633c9c6.jpg	ravendra	ravendra	0.5021879747348277	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\angle\ravendra\a35b0f62251f4773ae0569f71d118ba3.jpg	ravendra	ravendra	0.7436269069677914	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\lighting\ravendra\8b5221af4b654f01973d387baa1e5fce.jpg	ravendra	ravendra	0.7436269069677914	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\distance\ravendra\103c171bcf7340428ccf00f49a5cd24e.jpg	ravendra	ravendra	0.4727140671332853	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\distance\ravendra\143852b7e5764fe480997be906bb3cf7.jpg	ravendra	ravendra	0.5011308803343132	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\blur\ravendra\b82b2d3507cc4ab49d1d20b9c8dc7ef0.jpg	ravendra	ravendra	0.7736438315211082	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\multiple_faces\ravendra\bdf58eda61214032880fdee25686cd32.jpg	ravendra	ravendra	0.4668201071133028	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\normal\ravi\1ff748b0db1f4d9b88b4c5176386658d.jpg	ravi	ravi	0.0212383022159273	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\normal\ravi\93686ec7f91c418aa37c28912b134098.jpg	ravi	ravi	0.0376771157861199	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\angle\ravi\bd7f690c2bcd4fd39caa74042f0a16d8.jpg	ravi	ravi	0.0376771157861199	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\lighting\ravi\efb8397169914bf3aa31ecf01b687dae.jpg	ravi	ravi	0.4482681813914063	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\lighting\ravi\d364e0f3e65940a68528351029fe72db.jpg	ravi	ravi	0.4482681813914063	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\distance\ravi\02badeacfa984bf5b4c05262fb51f548.jpg	ravi	ravi	0.4482681813914063	ok	0.4	False
Facenet512	genuine	evaluation\dataset\victims\test\distance\ravi\8db138824f1d42b4a41945886f697eba.jpg	ravi	ravi	0.0265141268279222202	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\multiple_faces\ravi\041528dd6f1b4e68b65b9040f7309345.jpg	ravi	ravi	0.0376771157861199	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\normal\shubham\f34981c1e8ef4e5282426332e6fa0851.jpg	shubham	shubham	0.1404571915955 4215	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\angle\shubham\fcae35a9f88c4035b7f8da6a44bf1549.jpg	shubham	shubham	0.0089209889147 9387	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\distance\shubham\657bff67ea954cdc9cf4ef242a7bdb87.jpg	shubham	shubham	0.2917088813816 7916	ok	0.4	True
Facenet512	genuine	evaluation\dataset\victims\test\blur\shubham\4570bcdc073846029ac8515c481344dd.jpg	shubham	shubham	0.2461718841716 7113	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\normal\Maharsh\718c68ec86c845cc8e9b605220d6cbba.jpg	Maharsh	Maharsh	0.3441489536475387	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\angle\Maharsh\84f9319e32684b5fa35baf8c309885f1.jpg	Maharsh	Maharsh	0.2237377759445 0317	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ed6e345cfe0b4c1a91f7715f95cc59e3.jpg	Maharsh	Maharsh	0.3213541449207562	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\Maharsh\0d291f45643f4fecb868110f91ab344c.jpg	Maharsh	Maharsh	0.3213541449207562	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\Maharsh\f735fa5b36584eb08e9c957942b600e7.jpg	Maharsh	Maharsh	0.2567060095364 0443	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\blur\Maharsh\984206e6d13c4872bb4ef556aa27d5dd.jpg	Maharsh	Maharsh	0.2055746073306 5058	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0d4cfd9b1ea346d49bdab8e7acb5f6a5.jpg	Maharsh	Maharsh	0.5278478354639259	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\multiple_faces\Maharsh\0b43efbaa6a148a28ecf8fb4e07aa14c.jpg	Maharsh	Maharsh	0.2977065653741 1566	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\lighting\Maharsh\ba32ad437d5d43128b5e54de15a3d2a2.jpg	Maharsh	Maharsh	0.2977065653741 1566	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\lighting\Maharsh\71f33f70fda54b659e9f434e9700fd86.jpg	Maharsh	Maharsh	0.3492857634 36192	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\Maharsh\816dd95ad42740eb825d18b167afcc0b.jpg	Maharsh	Maharsh	0.2445740073047 2179	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\angle\Maharsh\6db477703c7a48fb927c05f75db4dc04.jpg	Maharsh	Maharsh	0.2297107137336034	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\Maharsh\89d1e95a4a8f4d598a9d3edfcf14489d.jpg	Maharsh	Maharsh	0.2791443317 99413	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\normal\ravendra\780bb17cf73a4a65a64cf9d19633c9c6.jpg	ravendra	ravendra	0.4288811384 6451625	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\angle\ravendra\a35b0f62251f4773ae0569f71d118ba3.jpg	ravendra	ravendra	0.6778060688495521	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\lighting\ravendra\8b5221af4b654f01973d387baa1e5fce.jpg	ravendra	ravendra	0.6778060688495521	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\distance\ravendra\103c171bcf7340428ccf00f49a5cd24e.jpg	ravendra	ravendra	0.4533187848668402	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\distance\ravendra\143852b7e5764fe480997be906bb3cf7.jpg	ravendra	ravendra	0.4084765568767 1414	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\blur\ravendra\b82b2d3507cc4ab49d1d20b9c8dc7ef0.jpg	ravendra	ravendra	0.7089994352559854	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\multiple_faces\ravendra\bdf58eda61214032880fdee25686cd32.jpg	ravendra	ravendra	0.4353303790836889	ok	0.4	False
ArcFace	genuine	evaluation\dataset\victims\test\normal\ravi\1ff748b0db1f4d9b88b4c5176386658d.jpg	ravi	ravi	0.0623827477 6383612	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\normal\ravi\93686ec7f91c418aa37c28912b134098.jpg	ravi	ravi	0.0990289092 5309404	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\angle\ravi\bd7f690c2bcd4fd39caa74042f0a16d8.jpg	ravi	ravi	0.0990289092 5309404	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\lighting\ravi\efb8397169914bf3aa31ecf01b687dae.jpg	ravi	ravi	0.3499237161385672	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\lighting\ravi\d364e0f3e65940a68528351029fe72db.jpg	ravi	ravi	0.3499237161385672	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\ravi\02badeacfa984bf5b4c05262fb51f548.jpg	ravi	ravi	0.3499237161385672	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\ravi\8db138824f1d42b4a41945886f697eba.jpg	ravi	ravi	0.0606340903592845 76	ok	0.4	True

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
ArcFace	genuine	evaluation\dataset\victims\test\multiple_faces\ravi\041528dd6f1b4e68b65b9040f7309345.jpg	ravi	ravi	0.09902890925309404	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\normal\shubham\f34981c1e8ef4e5282426332e6fa0851.jpg	shubham	shubham	0.2454647813549261	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\angle\shubham\fcae35a9f88c4035b7f8da6a44bf1549.jpg	shubham	shubham	0.0183253444947874	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\distance\shubham\657bff67ea954cdc9cf4ef242a7bdb87.jpg	shubham	shubham	0.3457354181 65714663	ok	0.4	True
ArcFace	genuine	evaluation\dataset\victims\test\blur\shubham\4570bcdc073846029ac8515c481344dd.jpg	shubham	shubham	0.5002346255170234	ok	0.4	False

C – Impostor Trial-Level Results

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	Maharsh	0.999286440608704	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravendra	1.004537108027414	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shivam	1.016702985713 7113	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravi	1.079689042 8853685	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shubham	0.980328190 4859983	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	Maharsh	0.732631322 0869628	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravendra	0.944543047 5266517	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shivam	0.774726708 3882409	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravi	0.937444821 0662871	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shubham	0.952789953 3769907	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	Maharsh	1.063191831 4214708	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravendra	0.983968302 7172427	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shivam	0.963982755 2521372	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravi	0.971825649 4447807	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shubham	1.136434883 3717421	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	Maharsh	0.588656526 7030961	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravendra	0.898554785 733229	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shivam	0.774410959 122571	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravi	0.671743054 5442693	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shubham	0.814417425 5313033	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	Maharsh	0.632475613 9154202	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	ravendra	0.841331744 4313957	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shivam	0.498566732 30472104	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	ravi	0.682211047 620376	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shubham	1.013739019 0846602	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	Maharsh	0.961804657 9070896	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravendra	1.087954701 6374382	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shivam	1.100369524 4547416	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravi	1.058211381 791922	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shubham	1.079920228 1246765	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	Maharsh	0.960760720 6425568	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravendra	1.030976530 3019673	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shivam	1.054833188 4255668	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravi	1.067561592 8456068	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shubham	1.218403213 116347	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	Maharsh	0.977808036 9886072	ok	0.4	False
Facene	impostor	evaluation\dataset\impostors\test\IMP_TE	IMP_TEST_00	ravendra	0.7570993150	ok	0.4	False

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
t		ST_007\image_08.jpg	7		003432			
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shivam	1.0318969386573504	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	ravi	0.9799067521093848	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shubham	0.9783127751777565	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	Maharsh	0.9355746252232577	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravendra	0.9180780450852638	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shivam	1.0857617101437202	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravi	0.8911268712591447	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shubham	0.8740102172952223	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	Maharsh	0.846474874303939	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravendra	0.9556919312208902	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shivam	1.0241293673665515	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravi	0.7943050836504475	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shubham	0.9817801939 04149	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	Maharsh	0.8253144447892153	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravendra	0.9081779999 67771	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shivam	0.9053691371 54513	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravi	1.1370217001 43828	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shubham	1.1041770005 51195	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	Maharsh	1.0090319090786066	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravendra	1.0880530110603583	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	shivam	1.0852905802 89451	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravi	1.0549709112350196	ok	0.4	False
Facenet	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	shubham	0.9672758844262591	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	Maharsh	0.7934343677293352	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravendra	0.9523720287904237	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shivam	1.2126203988345998	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravi	1.0989107388372759	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shubham	0.8751150540705973	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	Maharsh	0.8501426074205651	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravendra	0.8964914374275439	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shivam	0.7242675578526001	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravi	0.9037896437016473	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shubham	0.8958488457317231	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	Maharsh	1.1563923550165742	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravendra	1.0871287090244692	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shivam	1.1444387165117202	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravi	1.0464810804459888	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shubham	1.0276354256 20723	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	Maharsh	0.7895414352 37745	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravendra	1.0530975058379084	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shivam	0.7969624603301408	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravi	1.0243622211733674	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shubham	0.9991177956 31177	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	Maharsh	0.6545447417465784	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	ravendra	0.6916442894078731	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shivam	0.6226027578 09321	ok	0.4	False
Facene	impostor	evaluation\dataset\impostors\test\IMP_TE	IMP_TEST_00	ravi	0.7705933740	ok	0.4	False

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
t512		ST_005\image_05.jpg	5		811809			
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shubham	0.944145529261941	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	Maharsh	0.9778008846328943	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravendra	1.1361728650951926	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shivam	1.0661623888892495	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravi	1.1219908808006576	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shubham	1.2700593436504093	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	Maharsh	0.8482359286187355	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravendra	1.0399021472443	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shivam	1.0003006613310537	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravi	1.0026450255751975	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shubham	1.1581309782616358	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	Maharsh	0.925895902578914	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	ravendra	0.9785990743064846	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shivam	0.9355658436079676	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	ravi	0.9521509632447557	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shubham	0.9029385931774472	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	Maharsh	1.103603349629876	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravendra	0.9851707600205172	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shivam	1.0821574095438968	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravi	1.0782677700035552	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shubham	1.0230892462128822	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	Maharsh	0.8205401486115289	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravendra	0.8226821594212952	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shivam	0.8613308215535949	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravi	0.7847138749142214	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shubham	1.0950162167698065	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	Maharsh	0.8654026338681577	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravendra	0.8788618922322624	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shivam	0.7797544243 06893	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravi	0.9271983207707715	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shubham	1.0199411211757319	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	Maharsh	1.0691150844003037	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravendra	1.1063626266301783	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	shivam	1.1180013028686173	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravi	1.0748670383115566	ok	0.4	False
Facenet512	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	shubham	0.9213943534603892	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	Maharsh	0.8978200743301413	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravendra	0.8987874023026803	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shivam	1.0386211098802351	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	ravi	0.9328401609791679	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_001\image_01.jpg	IMP_TEST_001	shubham	0.8619149776 4827	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	Maharsh	0.9069707139368329	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravendra	1.0242950437610059	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shivam	0.7406550307176467	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	ravi	0.9523849332720967	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_002\image_02.jpg	IMP_TEST_002	shubham	0.9011396501140917	ok	0.4	False
ArcFac	impostor	evaluation\dataset\impostors\test\IMP_TE	IMP_TEST_00	Maharsh	0.8919941042	ok	0.4	False

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
e		ST_003\image_03.jpg	3		064368			
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravendra	1.1648178654813213	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shivam	0.8684743607521804	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	ravi	0.9812162753353975	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_003\image_03.jpg	IMP_TEST_003	shubham	0.9099681414865645	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	Maharsh	0.8799841879124635	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravendra	0.7889281152835895	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shivam	0.8642408526974453	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	ravi	0.961630408656887	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_004\image_04.jpg	IMP_TEST_004	shubham	0.9510954586166562	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	Maharsh	0.8331466759559374	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	ravendra	0.8616452491626548	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shivam	0.7944708472982027	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	ravi	0.8795488424996616	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_005\image_05.jpg	IMP_TEST_005	shubham	0.9800424355217633	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	Maharsh	0.7879155961422941	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravendra	1.0126676338483347	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shivam	0.8685754203628815	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	ravi	0.9043325881606347	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_06.jpg	IMP_TEST_006	shubham	0.8999506719925232	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	Maharsh	0.8497955770001818	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravendra	0.9583023928889242	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shivam	0.8089496240090243	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	ravi	0.9343719868031403	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_006\image_07.jpg	IMP_TEST_006	shubham	0.7633134003569432	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	Maharsh	0.8629072359731876	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	ravendra	0.9234958105053453	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shivam	0.92837586501532	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	ravi	0.9268847317372538	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_007\image_08.jpg	IMP_TEST_007	shubham	0.9654539507429147	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	Maharsh	0.8553226745622035	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravendra	0.9694625284749683	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shivam	0.8038722331172399	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	ravi	0.9015476862073633	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_09.jpg	IMP_TEST_008	shubham	0.8745768829231321	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	Maharsh	0.7169843985089579	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravendra	0.858815996667586	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shivam	0.7381320835620364	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	ravi	0.7524162385275535	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_008\image_10.jpg	IMP_TEST_008	shubham	0.9804495059932912	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	Maharsh	0.7902269851674018	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravendra	1.0107982857167543	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shivam	0.9491250322697444	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	ravi	0.9847303483317741	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_009\image_11.jpg	IMP_TEST_009	shubham	0.880192633337288	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	Maharsh	0.8370432488379553	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravendra	0.9298411028658115	ok	0.4	False
ArcFac	impostor	evaluation\dataset\impostors\test\IMP_TE	IMP_TEST_01	shivam	0.9660934093	ok	0.4	False

MODEL	TRIAL_TYPE	IMAGE_PATH	IMAGE_IDENTITY	TARGET_IDENTITY	DISTANCE	STATUS	DEFAULT_THRESHOLD	DECISION_AT_DEFAULT
e		ST_010\image_12.jpg	0		053438			
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	ravi	0.9252425350039606	ok	0.4	False
ArcFace	impostor	evaluation\dataset\impostors\test\IMP_TEST_010\image_12.jpg	IMP_TEST_010	shubham	0.816062803267116	ok	0.4	False

Threshold analysis

At threshold 0.40, all three models had zero FP in 60 negative trials. The threshold sweep shows the trade-off when the threshold is relaxed. For example, FaceNet recorded one FP at 0.50 and 0.55, and two FP at 0.60; at 0.60, FaceNet512 and ArcFace still recorded zero FP but their genuine acceptance counts did not automatically establish production superiority. Thresholds are model-specific and must be recalibrated on representative data.

Table A– Complete Threshold Comparison

MODEL	THRESHOLD	TP	TN	FP	FN	PRECISION	RECALL	F1	FAR	FRR	FALSE_VICTIM_MATCHES	GENUINE_TRIALS	IMPOSTOR_TRIALS
Facenet	0.2	15	60	0	17	1.0	0.46875	0.6382978723404256	0.0	0.53125	0	32	60
Facenet	0.25	17	60	0	15	1.0	0.53125	0.6938775510204082	0.0	0.46875	0	32	60
Facenet	0.3	17	60	0	15	1.0	0.53125	0.6938775510204082	0.0	0.46875	0	32	60
Facenet	0.35	19	60	0	13	1.0	0.59375	0.7450980392156863	0.0	0.40625	0	32	60
Facenet	0.4	21	60	0	11	1.0	0.65625	0.7924528301886793	0.0	0.34375	0	32	60
Facenet	0.45	24	60	0	8	1.0	0.75	0.8571428571428571	0.0	0.25	0	32	60
Facenet	0.5	26	59	1	6	0.9629629629629629	0.8125	0.8813559322033898	0.016666666666666666	0.1875	1	32	60
Facenet	0.55	32	59	1	0	0.9696969696969697	1.0	0.9846153846153847	0.016666666666666666	0.0	1	32	60
Facenet	0.6	32	58	2	0	0.9411764705882353	1.0	0.9696969696969697	0.03333333333333333	0.0	2	32	60
Facenet512	0.2	17	60	0	15	1.0	0.53125	0.6938775510204082	0.0	0.46875	0	32	60
Facenet512	0.25	18	60	0	14	1.0	0.5625	0.72	0.0	0.4375	0	32	60
Facenet512	0.3	19	60	0	13	1.0	0.59375	0.7450980392156863	0.0	0.40625	0	32	60
Facenet512	0.35	21	60	0	11	1.0	0.65625	0.7924528301886793	0.0	0.34375	0	32	60
Facenet512	0.4	22	60	0	10	1.0	0.6875	0.8148148148148148	0.0	0.3125	0	32	60
Facenet512	0.45	25	60	0	7	1.0	0.78125	0.8771929824561403	0.0	0.21875	0	32	60
Facenet512	0.5	27	60	0	5	1.0	0.84375	0.9152542372881356	0.0	0.15625	0	32	60
Facenet512	0.55	29	60	0	3	1.0	0.90625	0.9508196721311475	0.0	0.09375	0	32	60
Facenet512	0.6	29	60	0	3	1.0	0.90625	0.9508196721311475	0.0	0.09375	0	32	60
ArcFace	0.2	6	60	0	26	1.0	0.1875	0.3157894736842105	0.0	0.8125	0	32	60
ArcFace	0.25	11	60	0	21	1.0	0.34375	0.5116279069767442	0.0	0.65625	0	32	60
ArcFace	0.3	15	60	0	17	1.0	0.46875	0.6382978723404256	0.0	0.53125	0	32	60
ArcFace	0.35	23	60	0	9	1.0	0.71875	0.8363636363636363	0.0	0.28125	0	32	60
ArcFace	0.4	23	60	0	9	1.0	0.71875	0.8363636363636363	0.0	0.28125	0	32	60
ArcFace	0.45	26	60	0	6	1.0	0.8125	0.8965517241379310	0.0	0.1875	0	32	60
ArcFace	0.5	27	60	0	5	1.0	0.84375	0.9152542372881356	0.0	0.15625	0	32	60
ArcFace	0.55	29	60	0	3	1.0	0.90625	0.9508196721311475	0.0	0.09375	0	32	60
ArcFace	0.6	29	60	0	3	1.0	0.90625	0.9508196721311475	0.0	0.09375	0	32	60

Performance evidence

The benchmark performance file contains 55-image rows for each model. Average inference latency was approximately 3.5909591327 seconds for FaceNet, 3.9212489673 seconds for FaceNet512, and 3.8782448073 seconds for ArcFace in that benchmark run. The corresponding derived average FPS values were approximately 0.2784771315, 0.2550207876, and 0.2578486015. These are benchmark-run figures, not application-wide guarantees. The benchmark notes that inference latency includes detector, alignment/preprocessing, and embedding generation, and that GPU was not used.

Table A– Complete Performance Results

AVERAGE_COMPARISON_SECONDS	AVERAGE_EMBEDDING_GENERATION_SECONDS	AVERAGE_INFERENCE_LATENCY_SECONDS	CPU_RAM_GPU	DETECTOR_BACKEND	DISTANCE_METRIC	ERROR_IMAGES	FPS_FROM_AVERAGE_INFERENCE	IMAGE_COUNT	MEDIAN_INFERENCE_LATENCY_SECONDS	MODEL	NO_FACE_IMAGES	PREPROCESSING_SECONDS	STATUS
4.427500038139482e-05	3.590959132728114	3.590959132728114	CPU/RAM/GPU details in report; GPU not used	retinaface	cosine	0	0.2784771318854533	55	4.037023999990197	Facenet	0	included in inference latency; not separately exposed by DeepFace	measured
1.2485869734993447e-05	3.9212489672730126	3.9212489672730126	CPU/RAM/GPU details in report; GPU not used	retinaface	cosine	0	0.255020787597539	55	4.159344900006545	Facenet512	0	included in inference latency; not separately exposed by DeepFace	measured
1.2781521787553135e-05	3.8782448072729907	3.8782448072729907	CPU/RAM/GPU details in report; GPU not used	retinaface	cosine	0	0.25784860154384004	55	4.206378800008679	ArcFace	0	included in inference latency; not separately exposed by DeepFace	measured

Multi-frame evaluation

multi_frame_results.csv records not_available for all models and 1-, 3-, and 5-frame settings because no 44abelled video/frame sequences were present. Multi-frame 1/3/5 confirmation results are **Not measured in the current implementation/evaluation.** The current report therefore makes no claim that multi-frame confirmation improves accuracy. A future experiment must measure TP, TN, FP, FN, confirmation latency, and false Victim alerts at sequence level.

Table A – Complete Multi-Frame Evaluation Status

model	confirmation_frames	status	reason
Facenet	1	not_available	No labeled video/frame sequences in evaluation dataset.
Facenet	3	not_available	No labeled video/frame sequences in evaluation dataset.
Facenet	5	not_available	No labeled video/frame sequences in evaluation dataset.
Facenet512	1	not_available	No labeled video/frame sequences in evaluation dataset.
Facenet512	3	not_available	No labeled video/frame sequences in evaluation dataset.
Facenet512	5	not_available	No labeled video/frame sequences in evaluation dataset.
ArcFace	1	not_available	No labeled video/frame sequences in evaluation dataset.
ArcFace	3	not_available	No labeled video/frame sequences in evaluation dataset.
ArcFace	5	not_available	No labeled video/frame sequences in evaluation dataset.

14.6 Production versus benchmark configuration

MODE	MODEL	DETECTOR	METRIC	THRESHOLD	STATUS
Victim Search source default	FaceNet	OpenCV	Cosine	0.40	Current source default
Benchmark comparison	FaceNet	RetinaFace	Cosine	0.40	Measured benchmark row
Benchmark comparison	FaceNet512	RetinaFace	Cosine	0.40	Measured benchmark row
Benchmark comparison	ArcFace	RetinaFace	Cosine	0.40	Benchmark candidate
Staff Recognition	Configured model	Configured detector	Configured metric	Configured threshold	Staff-specific benchmark unavailable
Unknown Re-ID	Configured model/fallback	Configured detector	Configured metric	Local threshold map	Accuracy unavailable
Threat Detection	Separate heuristic	N/A	N/A	N/A	Face model is not used

Table A – Model Runtime Status Evidence

MODEL	BACKEND	METRIC	STATUS	EVIDENCE
Facenet	DeepFace	cosine	RUNTIME UNAVAILABLE	Labeled evaluation data is present, but the DeepFace/TensorFlow runtime could not be imported.
Facenet512	DeepFace	cosine	RUNTIME UNAVAILABLE	Labeled evaluation data is present, but the DeepFace/TensorFlow runtime could not be imported.
ArcFace	DeepFace	cosine	RUNTIME UNAVAILABLE	Labeled evaluation data is present, but the DeepFace/TensorFlow runtime could not be imported.
OpenCVFaceBackend	Haar + HOG/CLAHE fallback	cosine	ACTIVE BASELINE	OpenCV fallback is available; no independent recognition benchmark has been run.

Table B – Fallback Threshold Evaluation Status

THRESHOLD	METRIC	MODEL	STATUS	NOTES
0.2	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.25	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.3	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.35	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.4	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.45	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.
0.5	cosine	OpenCVFaceBackend	PLANNED	Threshold sweep is ready once dataset and runtime exist.

Interpretation

The benchmark shows that genuine acceptance changes with model and threshold. It does not measure live camera conditions, staff operation, unknown re-identification, or threat detection. The repository correctly preserves the production default separately from the candidate benchmark configuration

SOFTWARE TESTING

Test strategy

The project contains unit tests for the adapter, profile inventory, unknown-store behavior, Victim target filtering, browser annotation safety, dataset paths, metadata, and collector validation. The tests are primarily behavior and boundary tests; they are not a full browser end-to-end suite.

Representative functional test table

Test ID	Test case	Expected result	Evidence
T01	Missing authenticatio n secret	System remains unavailable and reports configuration error	load_auth_credentials()
T02	Legacy Member profile	Categorized as Staff	test_profile_labels_support_new_ and_legacy_prefixes
T03	Legacy Lost profile	Categorized as Victim	Same test
T04	Duplicate unknown DB rows	Count uses unique saved IDs	test_unknown_count_uses_unique _database_ids
T05	Clear unknown data	Images/cache/CSV tracking reset	test_clear_unknown_face_data_re moves_photos_and_resets_trackin g
T06	Victim cache restriction	Only selected Victim can match	test_known_face_cache_can_be_li mited_to_one_victim
T07	Five angle files	All saved angles enter cache	test_known_face_database_refres h_includes_all_saved_angles
T08	Browser unmatched Victim	Green detection box, no identity	browser annotation test
T09	Browser neural	Red selected-Victim result	neural runtime test

	selected Victim		
T10	Fallback Victim	No false identity label	fallback test
T11	Dataset path traversal input	Path remains under dataset root	collector test
T12	Enrollment face count	Exactly one face required	collector tests
T13	Test image face count	At least one face required; multiple allowed	collector tests

AI benchmark tests

The benchmark scripts validate model comparisons and threshold decisions using independent enrollment/test paths. The dataset metadata records conditions such as normal, angle, lighting, distance, blur, and multiple faces for Victim test images. The impostor source file records LFW identity/source references and hashes.

Test gaps

The repository does not contain a full Streamlit browser test, multi-user concurrency test, staff benchmark, unknown benchmark, threat benchmark, or labeled video sequence test. These gaps must remain visible in the submission.

Current validation status

The final audit attempted to execute the test suite with the project Python environments. The checked-in new_venv environment does not contain pytest, and the system Python 3.14 environment also reports No module named pytest; therefore a fresh test-run result cannot be claimed from this audit. A prior repository audit recorded 24 passed, 1 failed, and 1 error, but that result is historical evidence rather than a current rerun. The known assertion failure concerned the expected order of columns in evaluation/dataset_metadata.csv (file_path expected first by the test while the checked-in collector schema writes id first). The reported temporary-directory error was environment-related. These facts must be resolved or explicitly attached as test-run evidence before submission.

The evaluation folder also contains a documentation contradiction. The current measured result files and populated dataset are present, while older evaluation/README.md and evaluation/dataset/README.md text still describes the workspace as having no real evaluation data. The measured CSVs are used in Chapter 14 because they contain explicit values; the older status text is retained as legacy documentation and is not treated as a result.

SECURITY, PRIVACY, AND ETHICAL CONSIDERATIONS

Authentication and authorization

The system fails closed without configured credentials, compares secrets with hmac.compare_digest, and locks the current session after repeated failures. Administrator-only controls include registration, model configuration, and destructive local-data operations. Operators can monitor and review but cannot access those controls through the UI.

Password security limitations

The repository does not demonstrate password hashing, rotation, MFA, external identity federation, or a persistent login audit store. Environment variables and Streamlit Secrets are preferable to committed credentials, but deployment operators remain responsible for secret protection.

Data protection

Face images, unknown images, camera locations, and event histories are stored locally as ordinary image and CSV files. The source does not implement encryption at rest, retention expiry, file-level permissions, or a formal deletion approval system beyond administrator UI confirmation. A production deployment should add those controls.

Privacy and ethical use

Face recognition should be used only with appropriate authority, notice, consent or other lawful basis, and human review. A match distance is not a percentage certainty. A possible threat is not proof of a weapon. A locally assigned unknown ID is not a confirmed identity. The application should not be used as the sole basis for detention, disciplinary action, or emergency escalation.

Dataset licensing

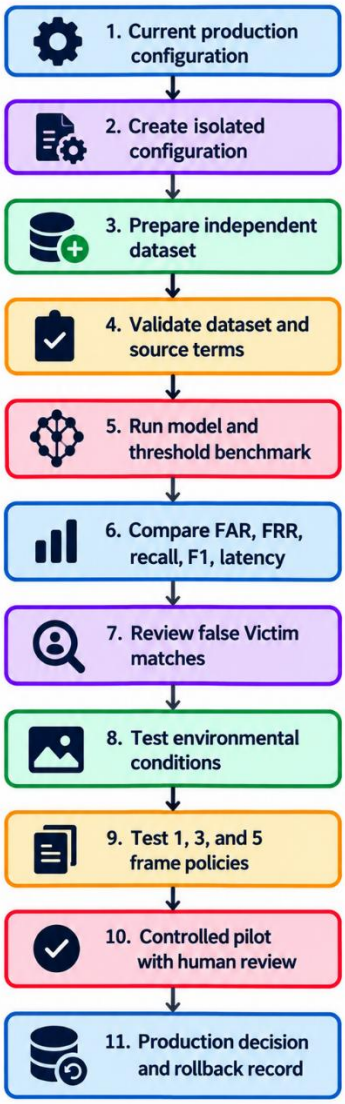
The impostor source file identifies LFW and preserves a source reference and terms note. Any final publication must include the required dataset attribution and check the applicable terms. The repository does not establish legal compliance with a specific jurisdiction.

LIMITATIONS AND FUTURE SCOPE

Current limitations

1. The benchmark is small and still-image based.
2. Staff-specific recognition accuracy is **Not measured in the current implementation/evaluation.**
3. Unknown Re-ID accuracy is **Not measured in the current implementation/evaluation.**
4. Threat-detection accuracy is **Not measured in the current implementation/evaluation.**
5. Multi-frame metrics are unavailable.
6. The active fallback is not a trained neural identity model.
7. Lighting, pose, blur, distance, occlusion, detector errors, and multiple faces remain challenging.
8. WebRTC worker annotation does not perform the same CSV/session-state writes as the OpenCV path.
9. File-based storage is not designed for concurrent distributed production use.
10. Application RAM, CPU, GPU usage, maximum camera count, and operational FPS are **Not measured in the current implementation/evaluation.**
11. Password hashing, encryption, TLS, retention, and comprehensive security auditing are not implemented.
12. Threat labels are heuristic and require human review.

Safe model-change workflow



The current production configuration must be recorded before a change. Enrollment and test identities must remain separated. The candidate configuration must be evaluated with independent impostors, actual deployment conditions, detector availability, latency, and resource cost. The change should be approved only after false Victim cases have been reviewed.

Future work

- Add larger and more representative Victim, Staff, Unknown, and threat datasets.
- Measure Staff recognition and Unknown Re-ID independently.
- Add labeled video sequences and evaluate 1-, 3-, and 5-frame confirmation.
- Calibrate thresholds by model, detector, camera, and environment.
- Add a database or transactional event store for concurrent operation.
- Add password hashing or an external identity provider.
- Add encryption, retention, consent, audit, and incident-response controls.
- Add reproducible application FPS, latency, CPU, RAM, GPU, and scale tests.
- Add automated browser tests and controlled deployment validation.
- Evaluate a trained threat detector separately from face recognition.
- Introduce multi-camera orchestration only after single-camera behavior is measured.

PREVIEW & SCREENSHOTS

Authentication

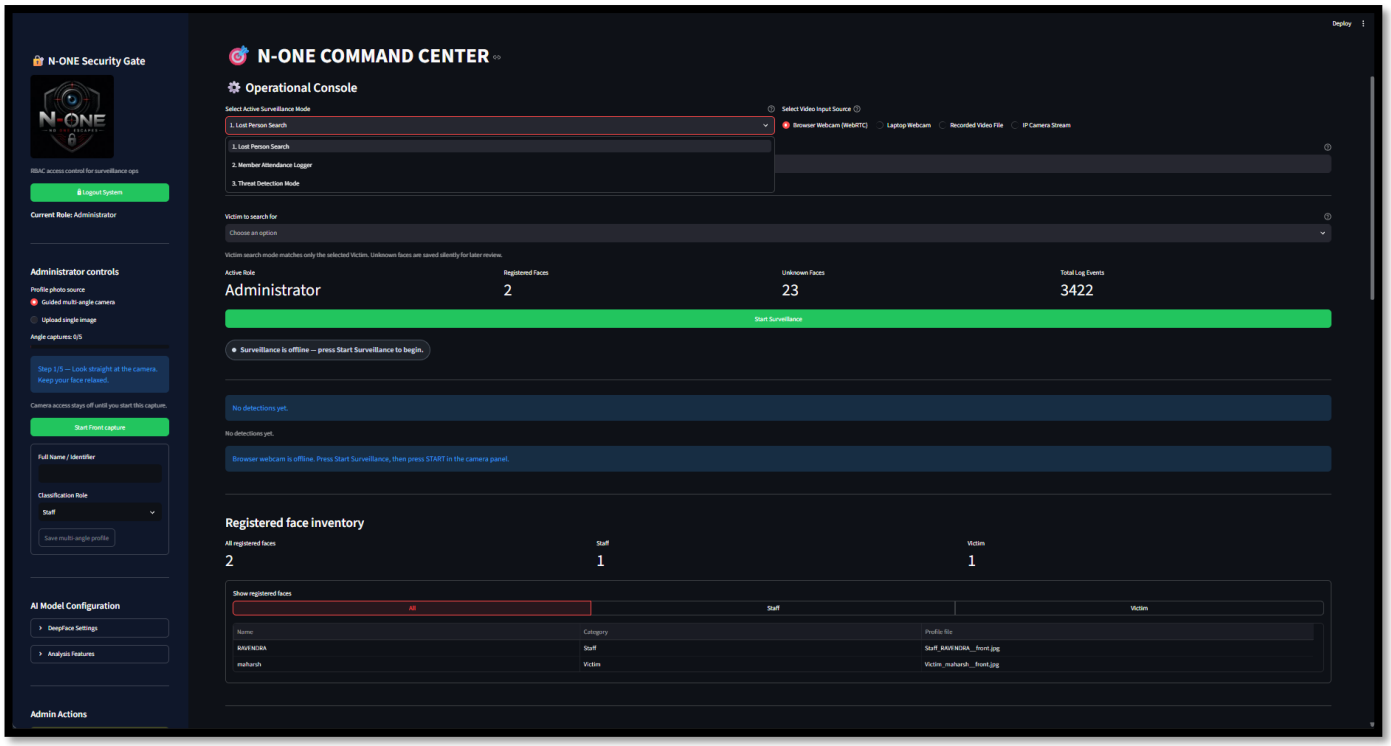
- **Authentication Fallback**



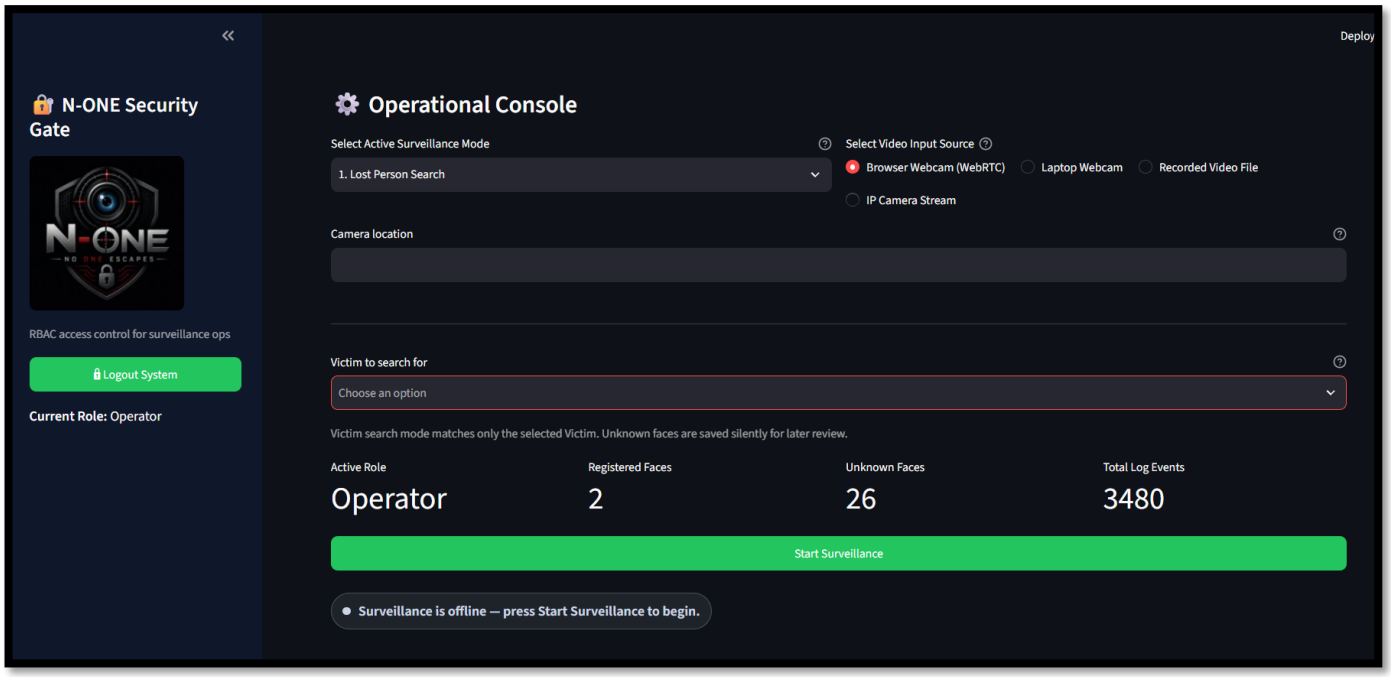
- **Invalid Login**



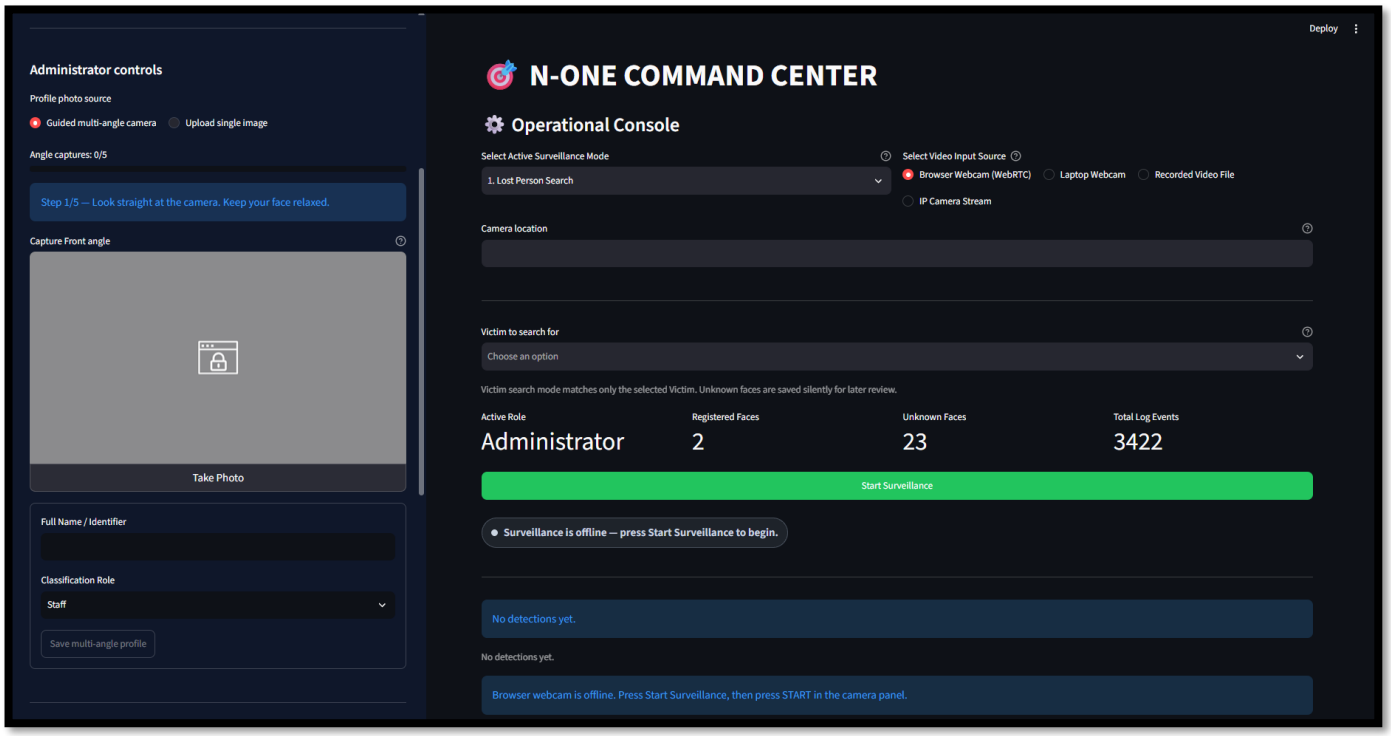
Admin Dashboard



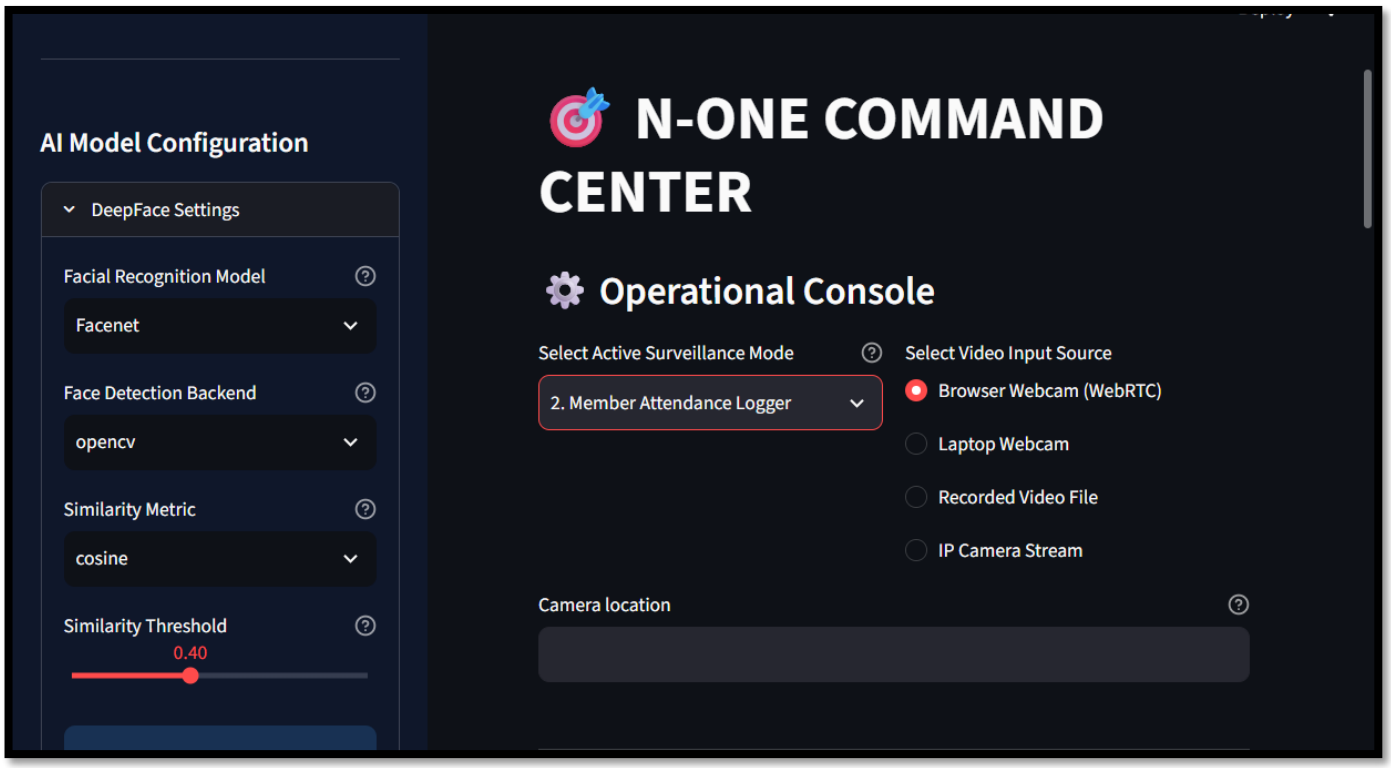
Operator Dashboard



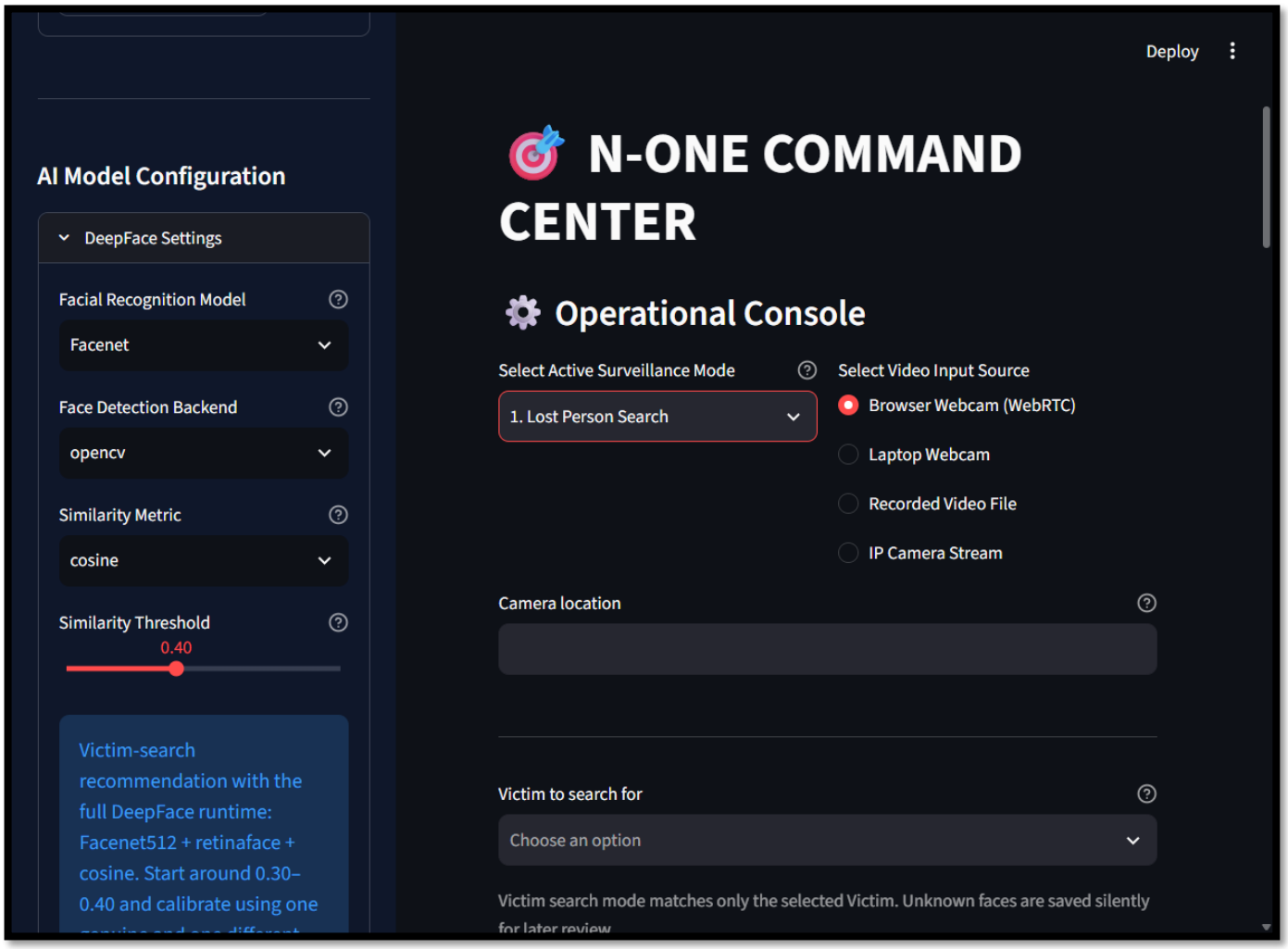
Profile Registration



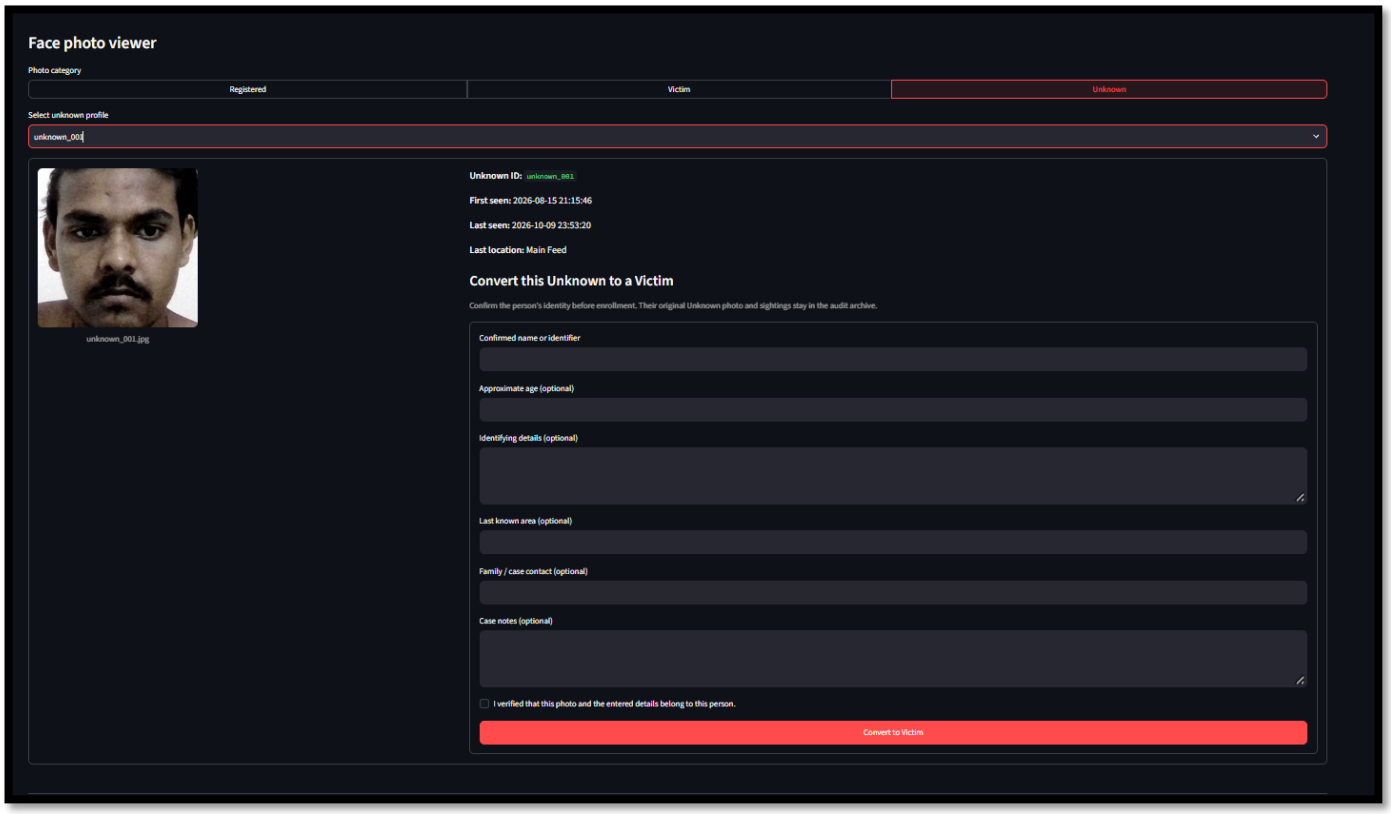
staff or Known



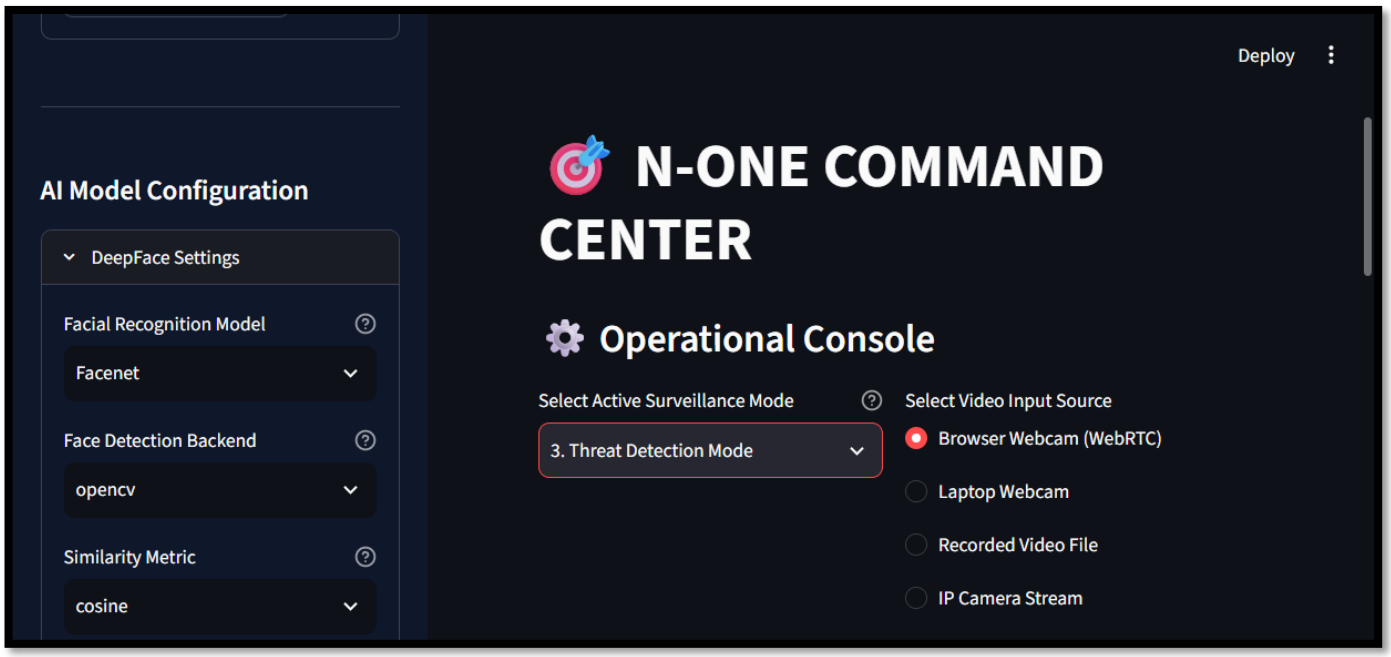
Victim Search



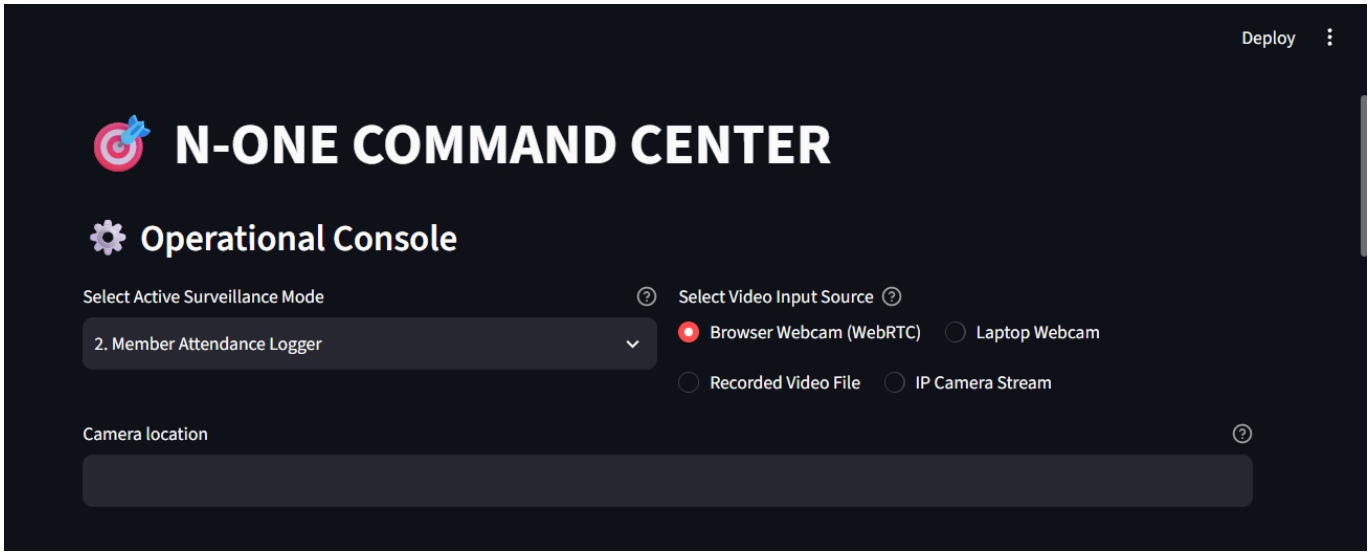
Unknown Reid



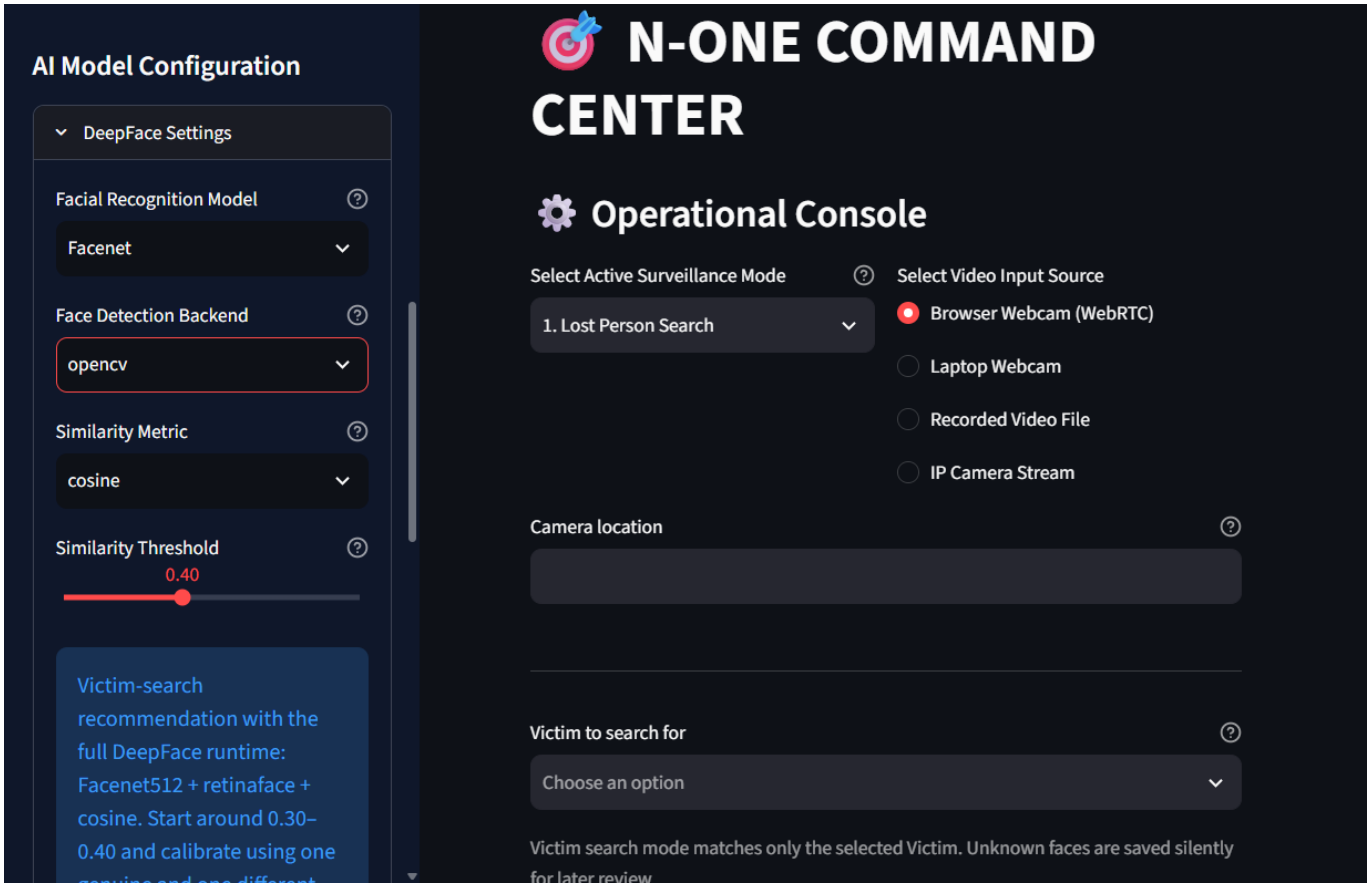
Threat Detection



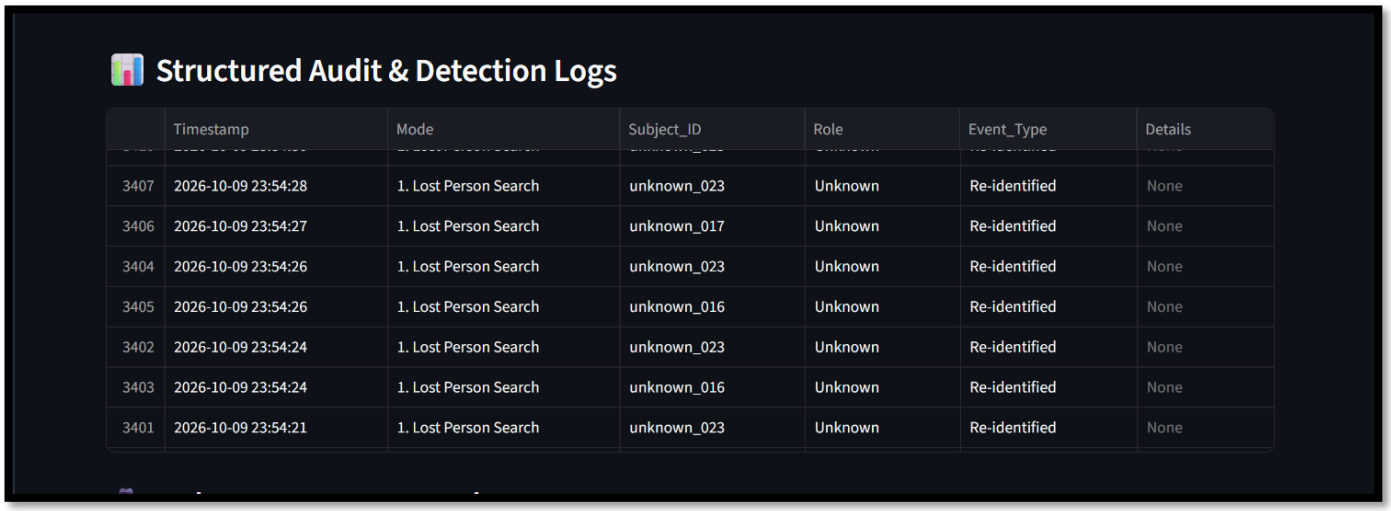
Camera Sources



Model Configuration



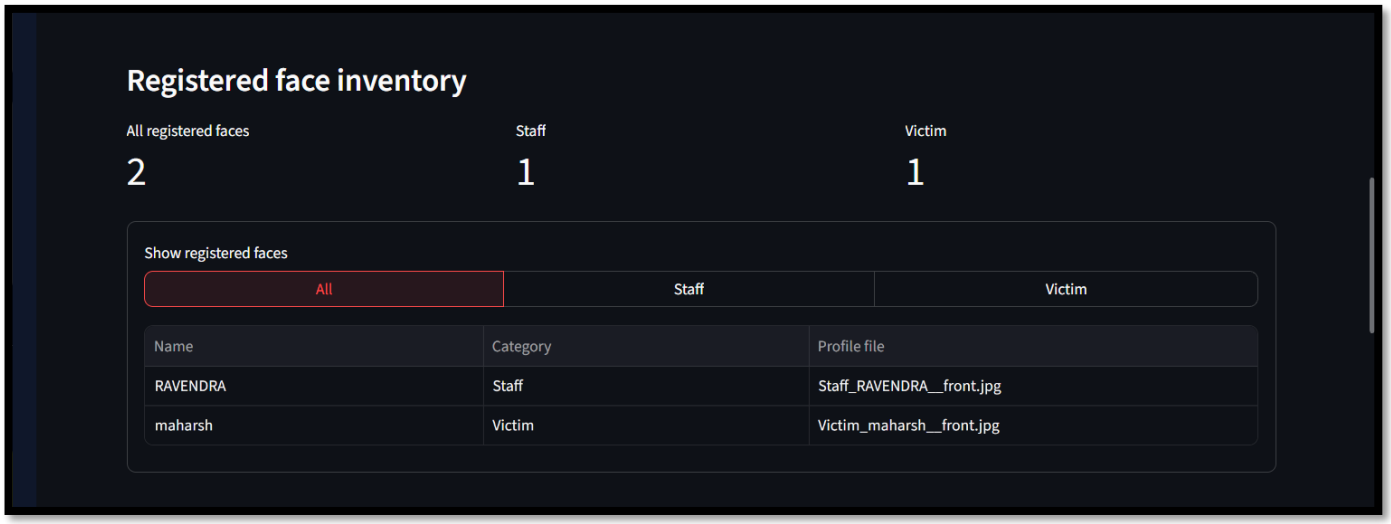
Logs And Audits



Structured Audit & Detection Logs

	Timestamp	Mode	Subject_ID	Role	Event_Type	Details
3407	2026-10-09 23:54:28	1. Lost Person Search	unknown_023	Unknown	Re-identified	None
3406	2026-10-09 23:54:27	1. Lost Person Search	unknown_017	Unknown	Re-identified	None
3404	2026-10-09 23:54:26	1. Lost Person Search	unknown_023	Unknown	Re-identified	None
3405	2026-10-09 23:54:26	1. Lost Person Search	unknown_016	Unknown	Re-identified	None
3402	2026-10-09 23:54:24	1. Lost Person Search	unknown_023	Unknown	Re-identified	None
3403	2026-10-09 23:54:24	1. Lost Person Search	unknown_016	Unknown	Re-identified	None
3401	2026-10-09 23:54:21	1. Lost Person Search	unknown_023	Unknown	Re-identified	None

Inventory



Registered face inventory

All registered faces: 2

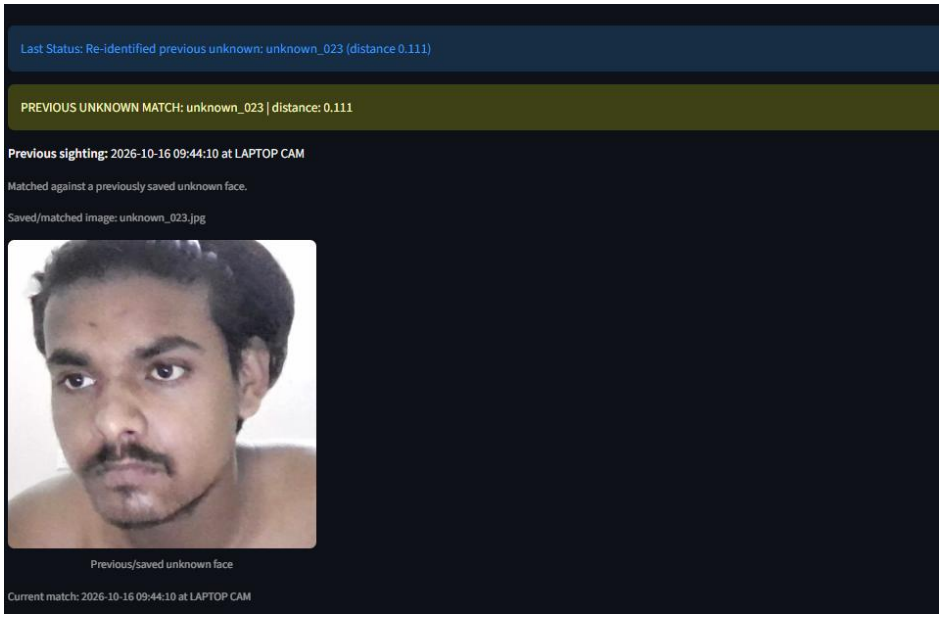
Staff: 1

Victim: 1

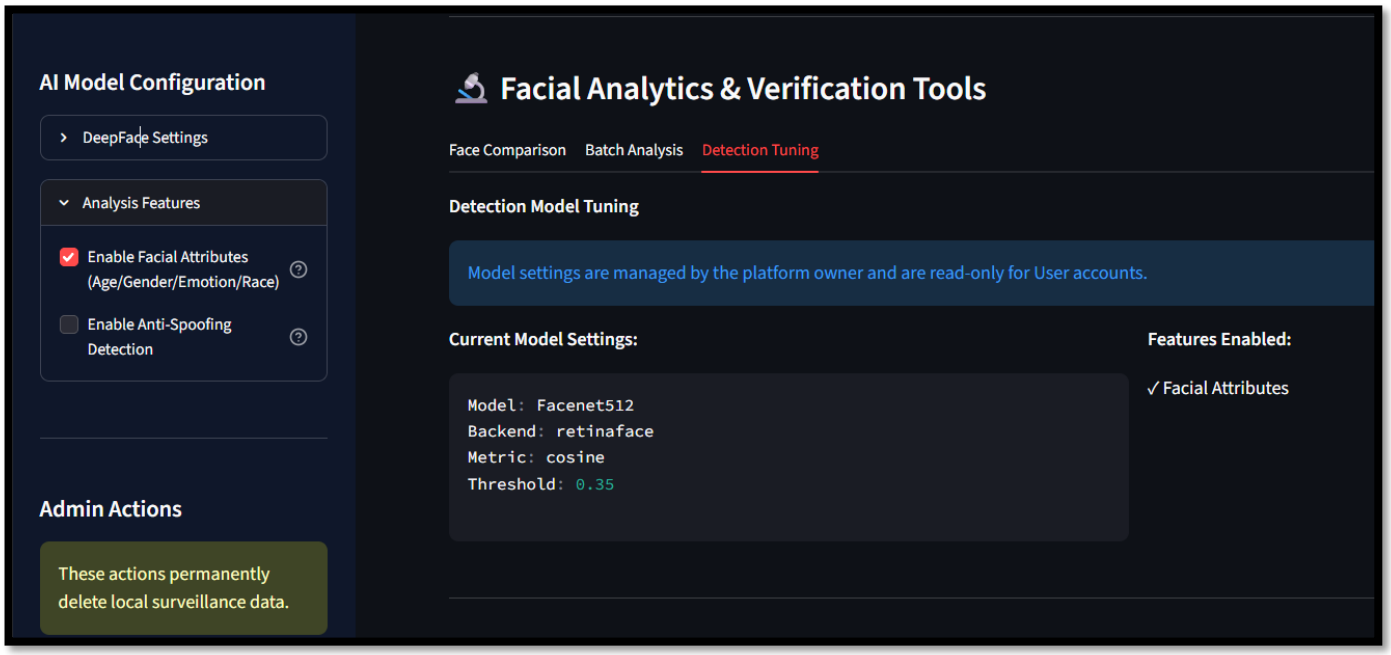
Show registered faces: All | Staff | Victim

Name	Category	Profile file
RAVENDRA	Staff	Staff_RAVENDRA__front.jpg
maharsh	Victim	Victim_maharsh__front.jpg

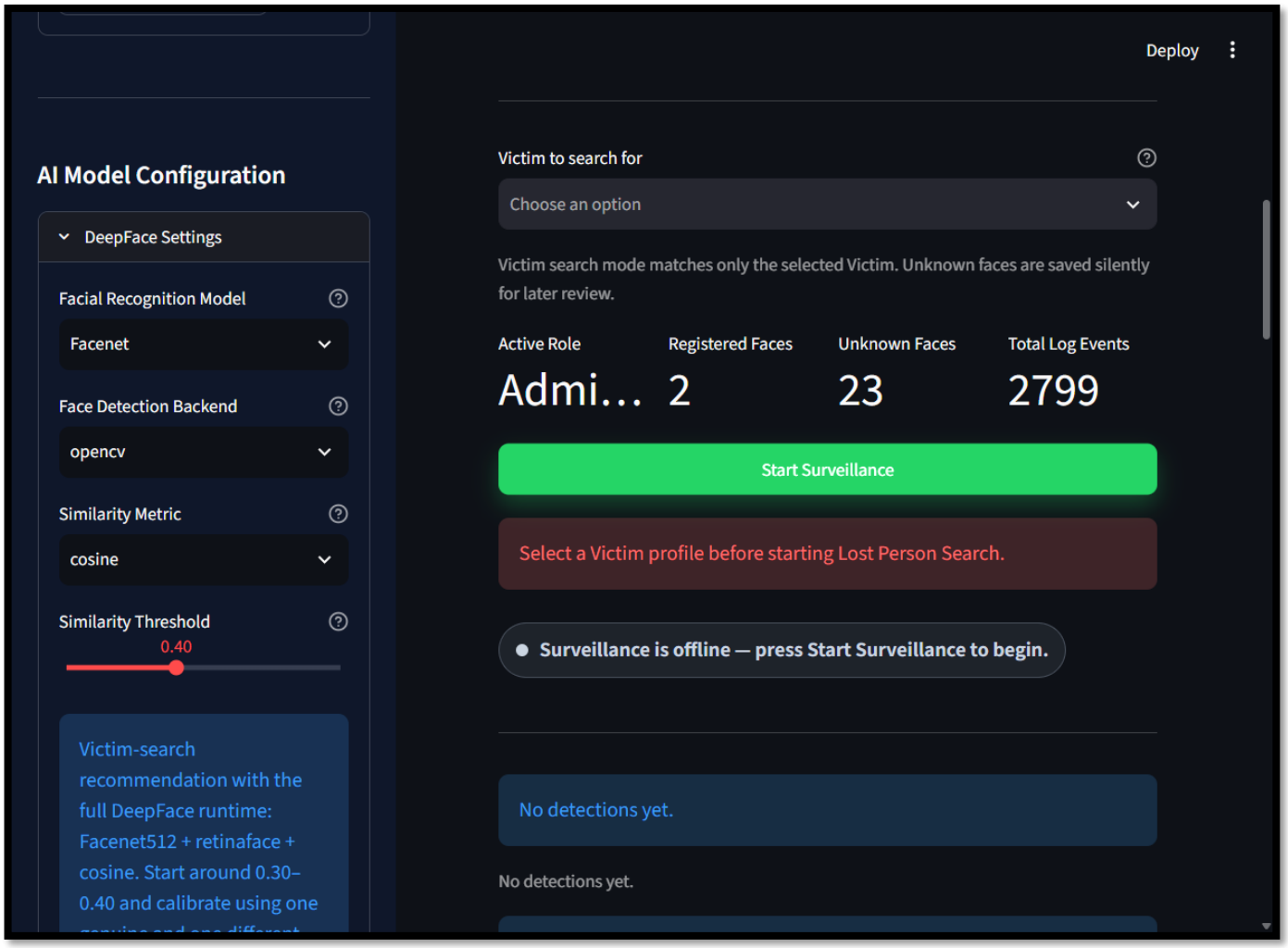
Results



Evaluation



Error Validation



CONCLUSION

N-ONE is an implemented Streamlit prototype that combines authenticated operation, face-only profile enrollment, Victim Search, registered-person recognition, local unknown-person re-identification, heuristic threat monitoring, and CSV/image evidence storage. Its strongest design decision is the explicit separation of tasks: Victim Search is a selected known identity workflow, Unknown Re-ID is a local same-unknown workflow, and Threat Detection is independent of face recognition.

The repository contains meaningful benchmark evidence. At threshold 0.40, ArcFace with RetinaFace and cosine distance produced the highest measured recall and F1 among FaceNet, FaceNet512, and ArcFace in the limited evaluation sample, with zero false Victim matches in the tested negative trials. The live source default remains FaceNet with OpenCV, cosine distance, and threshold 0.40, and the audited lightweight runtime uses an OpenCV/HOG fallback when TensorFlow is unavailable. These facts must remain distinct.

The project is not a claim of universal accuracy or guaranteed safety. Staff, Unknown Re-ID, threat, and multi-frame performance remain unmeasured. Security and privacy controls require strengthening before production use. The appropriate next step is a controlled evaluation program with larger independent data, sequence-level validation, human review, stronger storage and identity controls, and reproducible deployment measurements.

BIBLIOGRAPHY / REFERENCES

1. Schroff, F., Kalenichenko, D., and Philbin, J. (2015). *FaceNet: A Unified Embedding for Face Recognition and Clustering*. Proceedings of CVPR.
2. Deng, J., Guo, J., Xue, N., and Zafeiriou, S. (2019). *ArcFace: Additive Angular Margin Loss for Deep Face Recognition*. Proceedings of CVPR.
3. OpenCV Documentation. *Cascade Classifier and Computer Vision Library Documentation*. https://docs.opencv.org/
4. Streamlit Documentation. *Streamlit Documentation*. https://docs.streamlit.io/
5. DeepFace project documentation and source tree included under the repository's deepface/ directory. The project uses it conditionally through deepface_adapter.py.
6. Labeled Faces in the Wild source reference recorded in evaluation/impostor_sources.csv. The applicable dataset terms must be reviewed before publication.
7. N-ONE internal documentation: README.md, BRAIN.md, ARCHITECTURE.md, AI_MODEL_CONFIGURATION.md, docs/AI_MODEL_EVALUATION_REPORT.md, and docs/AI_MODELS_AND_CONFIGURATION_GUIDE.md.
8. N-ONE source and tests: app.py, deepface_adapter.py, evaluation/scripts/, and tests/.